



**SHOULD REAL ESTATE APPRECIATION BE INCORPORATED INTO
VAN-ES CALCULATIONS FOR URBAN TRANSIT INVESTMENTS?
EVIDENCE FROM MONTPELLIER'S INFRASTRUCTURE PROJECTS**

Master Thesis

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“All models are wrong but some are useful”

(Box, 1979, p. 202)

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Abstract

The socio-economic net present value (VAN-ES) is the metric through which French public authorities, and the citizens consulted in enquêtes publiques, judge whether a major infrastructure project is worth its cost. Yet it omits one of the largest economic effects transit produces: the capitalization of the investment into surrounding property values. Using difference-in-differences on DVF transaction data (2015–2025) for two contemporaneous Montpellier projects — tramway Line 5 and the Bustram A bus-rapid-transit corridor — this thesis estimates that properties within 500 m of a Line 5 station appreciated roughly €174/m² (4.58%) more than comparable controls once construction began, a residential capitalization on the order of the line's €440 million construction cost and absent from its VAN-ES. The effect tracks the credibility of the public commitment rather than transport mode alone, is largest in lower-income neighborhoods, and depends on accompanying redevelopment. Incorporating a mode-specific, credibility-timed real estate parameter would make the metric reward integrated transit-and-land-use planning — and, as much to the point, give decision-makers and the public a fuller account of the value a project creates and of who captures it.

Keywords: socio-economic evaluation (VAN-ES); transit capitalization; difference-in-differences; real estate value; transit-oriented development

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1. Introduction

The Valeur Actuelle Nette Économique et Sociale (VAN-ES) is the standard decision-making tool used by French public entities to evaluate major infrastructure projects (Instruction du Gouvernement, 2014; Quinet, 2013). As prescribed by CEREMA (CEREMA, 2021) guidelines, this methodology extends traditional VAN (Net Present Value) analysis by incorporating broader social and environmental externalities, enabling decision-makers to compare projects based on their expected outcomes for both the public entity and the community. Despite encompassing a wide range of parameters — from travel time savings to environmental impacts, VAN-ES systematically omits one significant effect: capitalization of infrastructure investments into property values and its consequences on residential cost (Alonso, 1964; Muth, 1969; Mills, 1972) and municipal tax revenues.

Empirical evidence suggests that urban transit investments affect real estate markets, with properties near high-quality public transport — defined by frequency, reliability, safety, and sufficient capacity (CEREMA, 2021, p. 15) — public transport resulting in price premiums that reflect improved transportation flows (Debrezion, Pels & Rietveld, 2007, IAU-IFSTTAR, 2011). These price effects are not transfers of value but represent real economic gains: reduced residential costs for households who can substitute private vehicle ownership, increased density enabling more efficient land use, and enhanced municipal tax bases through higher property value (Quinet, 2013 and Oates & Schwab, 2009). This analysis is all the more relevant as constrained public finances and competing demands on municipal budgets make efficiency ever more important in public spending decisions (Loi n° 2012-1558, 2012). Finally, France's Zéro Artificialisation Nette objectives incentivize densification over urban sprawl making land appreciation due to urban redevelopment even more relevant (Loi n° 2021-1104, 2021).

This paper examines whether real estate appreciation could and should be incorporated into VAN-ES calculations for urban transit investments. Using Montpellier Méditerranée Métropole's recently completed tramway Line 5 and the Bustram A high-frequency Bus as a case study and using a difference-in-differences econometric analysis on property transaction data this paper aims at isolating the causal effect of the new line on surrounding land values. Then, demonstrate the impact of such an addition on existing VAN-ES frameworks and quantify the difference this adjustment would have made in project evaluation. That said, this paper does not intend to provide urban planners with a specific methodology for forecasting urban appreciation into VAN-ES. Instead, it focuses on demonstrating the relevance of such a

measure. The rest of this thesis will proceed as follows: Section 2 reviews the current VAN-ES methodology as applied in France, the theoretical and empirical literature on transit capitalization into property values, and the policy debate surrounding value capture mechanisms. Section 3 presents the data sources for this research — primarily the Demandes de Valeurs Foncières (DVF) database — and the econometric methodology for estimating causal property value effects. Section 4 reports our empirical results for Line 5 and the Bustram A, simulates an adjusted VAN-ES calculation, and discusses the implications for infrastructure planning and public finance. This paper then concludes with policy recommendations and avenues for further research.

2. Literature Review

This literature review delves into three interconnected research areas crucial to this analysis. Firstly, it examines the current VAN-ES methodology applied in France to establish the foundational framework for the paper's proposed extension. Secondly, it reviews and contextualizes a portion of the extensive literature on transit capitalization and its impact on property values. This provides the theoretical and empirical basis for this contribution. Lastly, the paper focuses on the policy debate surrounding value capture mechanisms that aim to link empirical findings to broader questions of public finance and infrastructure funding.

2.1 The VAN-ES Framework in France

2.1.1 Regulatory Context and Evolution

The socio-economic evaluation of public investments in France mostly comes from a long tradition of cost-benefit analysis, formalized through successive methodological refinements (Cerema, 2021). The mandate to produce a socio-economic evaluation for major transport infrastructure projects was introduced in 1982 by Article 14 of the Loi d'Orientation des Transports Intérieurs (Loi n° 82-1153, 1982). This requirement was subsequently strengthened by the 2012 programming law on public finances, which mandates evaluation for all civil investment projects financed by the State above a threshold of approximately €83 million (Loi n° 2012-1558, 2012).

The current methodological framework derives primarily from the 2014 government instruction, itself building upon the foundational reports by Boiteux (1994, 2001), Lebègue (2005), and Quinet (2013). The Quinet report, produced under the supervision of the Commissariat Général à la Stratégie et à la Prospective, represents the most comprehensive recent synthesis and established the parameters still in use today.

2.1.2 Structure of VAN-ES Evaluation

CEREMA's guidelines define socio-economic evaluation using three principal components, as follows.

Strategic Analysis (Analyse Stratégique). This component reviews the territorial context, existing transport supply and demand, the reference scenario (what would occur without the project), and the project's objectives. It situates the investment within a broader set of planning goals and identifies the problems it aims to address.

Effects Analysis (Analyse des Effets). This quantifies and, where possible, monetizes the project's

impacts across multiple dimensions: travel time savings, operating cost changes, safety improvements, environmental effects (air quality, noise, greenhouse gas emissions), and effects on urbanism. Some effects cannot be assessed quantitatively and are assessed qualitatively or through multi-criteria analysis.

Synthesis and Indicators. The evaluation culminates in the calculation of several key indicators, including the Valeur Actualisée Nette Socio-Économique (VAN-ES), that represents the present value of net benefits to society. Additional mandatory indicators include the VAN-ES per euro invested, the VAN-ES per euro of public expenditure, the internal rate of return, and the optimal commissioning date.

2.1.3 Discount Rate and Valuation Parameters

The discount rate plays a critical role in VAN-ES calculations as it determines the relative weight given to future versus present costs and benefits. Quinet (2013) initially recommended a risk-free discount rate of 2.5%, with a risk premium of 2% multiplied by the project's socio-economic beta (β). This framework was subsequently revised by France Stratégie (2021), which lowered the risk-free rate to 1.2%, yielding — when β is assumed equal to 1 — a composite discount rate of 3.2%. This rate has declined substantially from historical levels: the French discount rate was 8% until 2005, then 4% under the Lebègue (2005) recommendations, reflecting both lower real interest rates and longer expected benefit horizons from infrastructure projects. To illustrate the practical significance: at an 8% discount rate, a benefit occurring in 30 years is worth only about 10% of the same benefit today; at 3%, it is worth about 40%.

It is also worth mentioning that the socio-economic beta should not be confused with the financial beta used in corporate finance in the Capital Asset Pricing Model (by Sharpe, 1964). Here, β measures the correlation between the project's expected benefits and aggregate economic growth. Time savings from transport projects depend on traffic volumes — themselves correlated with economic activity — which warrants a higher discount rate than a flood protection investment whose benefits are independent of GDP. The default assumption of $\beta = 1$ for transport projects is recommended by Quinet (2013) when the project-specific beta has not been estimated. It reflects the general expectation that transport benefits co-move positively with GDP growth. However, it is more a simplifying convention rather than an empirically calibrated parameter. This convention may nonetheless become harder to sustain going forward: incorporating real estate appreciation into the VAN-ES ties the benefit stream more directly to property-market cyclicalities, which is typically more volatile and more sensitive to the economic cycle than aggregate time savings. This strengthens the case for estimating a project-specific

β — one that may lie closer to the β of the real estate sector — rather than defaulting to $\beta = 1$.

Additionally, the 2021 calibration was conducted during a period of historically low interest rates, with the European Central Bank's deposit facility rate at 0%. Since then, the monetary environment has shifted considerably: the ECB raised its key rate to 4.5% by September 2023 before beginning a gradual easing cycle in 2024 (European Central Bank, 2024). This creates a tension. The socio-economic discount rate is not intended to track market rates mechanically but instead, reflects the society's preference for present over future and incorporates intergenerational equity considerations that differ from financial discounting. Nevertheless, the gap between the recommended 1.2% risk-free rate and prevailing market rates raises questions about whether a further revision may be warranted. A higher discount rate would reduce the present value of long-term benefits, making it harder for infrastructure projects to pass the VAN-ES threshold — and conversely making the inclusion of near-term real estate appreciation effects, which materialize within a few years of construction rather than over decades, relatively more important in the overall evaluation, therefore supporting the idea of adding real estate appreciation into VAN-ES computation.

Additionally, expenditures financed through public funds are multiplied by a coefficient of 1.2 to account for the distortionary costs of taxation (the coût d'opportunité des fonds publics). This shadow price of public funds may itself warrant reassessment in a context of tighter fiscal constraints and rising debt servicing costs across European sovereigns. Finally, unit values for non-market effects — the value of time, the value of a statistical life, the social cost of carbon — are periodically updated through expert commissions. The Quinet (2013) report provides the current reference values, though these are acknowledged to carry substantial uncertainty.

2.1.4 What VAN-ES Currently Excludes

Despite its comprehensive ambition, the VAN-ES framework systematically omits several categories of effects. Most relevant to our analysis, real estate price appreciation attributable to improved transportation is not included in standard calculations. The CEREMA methodology focuses on direct user benefits (travel time, vehicle operating costs) and external effects (pollution, safety) but does not attempt to capture the capitalization of these benefits into land values. This omission is motivated by both practical and theoretical considerations. Practically, measuring property value effects requires econometric analysis that goes beyond standard traffic modeling as it is often difficult to separate the effect attributable to an infrastructure project. Theoretically, there has been debate about whether property value changes represent « new » benefits or transfers that are already captured elsewhere in the calculation (a point I will address in Section 2). Similarly, are excluded from the calculation

induced economic development effects beyond direct construction impacts; changes in municipal tax revenues; and long-term effects on urban form and density. The framework explicitly warns against double-counting, noting that some apparent benefits may simply reflect redistribution rather than net social gains.

2.2. Transit Capitalization into Property Values

2.2.1 Theoretical Foundations

The relationship between transport accessibility and land values has been recognized since the nineteenth century. Von Thünen (1826) proposed an agricultural location model demonstrating that land rent decreases with distance from a central market as transport costs increase. This insight was later formalized in the urban economics literature by Alonso (1964), who adapted the model to urban contexts through bid-rent theory; Muth (1969), who extended the framework to explain residential density patterns; and Mills (1972), who further developed the monocentric city model (all discussed in Brueckner, 2011). The core insight is that households and firms bid for locations based on the total cost of occupying that location, which includes both rent and transport costs. Improvements in accessibility reduce transport costs, thus enabling occupants to bid higher rents while maintaining the same total expenditure, therefore capitalizing accessibility gains into property values.

Rosen (1974) provided the econometric framework for recovering these implicit prices through hedonic regression, a revealed preference method that decomposes the price of a differentiated good into the contributory values of its constituent characteristics. In this approach, property prices are modeled as a function of structural characteristics (size, age, quality), neighborhood attributes, and accessibility measures. The coefficient on an accessibility variable — such as distance to a transit station — extracts the marginal willingness to pay for that service.

For transit specifically, the accessibility premium reflects several components such as reduced commute time and cost, savings induced by reduced need for automobile ownership, expanded access to the urban area — thus improving employment opportunities — and amenity benefits from transit-oriented neighborhood characteristics. These benefits vary by transit mode, with higher-capacity, higher-frequency services (metro, light rail) generally commanding larger premiums than lower-capacity modes (bus).

That said, it is important to nuance these benefits as they can be perceived as detrimental for specific populations (thus reducing their bid for high-value properties or the offer on low-value ones). By connecting areas that were previously separated by commute time, transit infrastructure effectively

merges distinct urban environments into a unified accessible space, thus bringing together neighborhoods with different socioeconomic profiles, housing typologies, and ways of life. This spatial integration, while economically and socially efficient at the aggregate level, can trigger two opposing dynamics that both generate resistance at micro-level.

On one hand, improved accessibility to lower-income neighborhoods can accelerate gentrification: as wealthier households are attracted by the new connectivity, property values and rents increase, gradually displacing incumbent residents who can no longer afford to live in their own neighborhood. Grube-Cavers and Patterson (2015) documented this pattern across Canadian cities, finding that new rapid transit stations were associated with significant increases in neighborhood income and education levels, both indicators of gentrification. Bardaka et al. (2018) similarly found that light rail investments in the United States disproportionately increased property values in lower-income station areas, raising concerns about residential displacement. In Montpellier's case, several neighborhoods along Line 5's western section — including Estanove, Chamberte, and the Pas-du-Loup — are classified as *quartiers prioritaires de la politique de la ville*, making this concern directly relevant to our study area. This tension is not new in Montpellier: since the 1990s, urban planners — notably under the leadership of Mayor Georges Frêche — deliberately designed tramway alignments to traverse both affluent and lower-income neighborhoods on the same lines, using transit as an instrument of social mixing (Douay, 2009). Line 1 connects the city center to the quartier de la Mosson, one of Montpellier's most prominent social housing districts; Line 3 links the Écusson to the peripheral communes of Lattes and Juvignac while passing through the Pergola and Cévennes neighborhoods. Line 5 continues this tradition by connecting Clapiers and the university district to the western *quartiers prioritaires*, explicitly extending the policy of using infrastructure as a lever for territorial cohesion.

On the other hand, residents of quieter, more affluent neighborhoods may perceive the arrival of transit as a threat to their quality of life as they may express fear of increased noise, foot traffic, and the proximity of populations from different social backgrounds. Residents of Lavérune, for instance, explicitly opposed the extension of Line 5 into their commune, rejecting any densification between the village and Montpellier's urban fringe. This form of resistance, sometimes characterized as NIMBYism (Not In My Back Yard), reflects a perceived degradation of neighborhood exclusivity that transit connectivity can induce (Dear, 1992). Kahn (2007) showed that the direction of the effect — whether transit raises or lowers nearby property values — depends critically on the pre-existing socioeconomic character of the neighborhood: transit increases values in poorer areas (via accessibility gains) but can decrease values in wealthier areas for certain type of assets via “unwanted” social mixing.

A related but distinct mechanism operates at the property level. Transit-induced densification can

create a divergence between land value and property value. When zoning changes or market expectations shift toward higher-density development, the land beneath a large, low-density house may appreciate while the house itself loses value as its current form no longer represents the "highest and best use" of the plot (Arnott & Lewis, 1979). Property owners who are unwilling or unable to subdivide their land and redevelop at higher density may find themselves holding an asset whose market price falls below what the land alone would command. This paradox — rising land values coexisting with declining property values — is a well-documented consequence of transit-oriented densification and represents a real cost to incumbent homeowners that aggregate VAN-ES calculations do not capture.

These distributional tensions are entirely absent from VAN-ES methodology, which aggregates costs and benefits across all affected populations without distinguishing winners from losers. Yet they are central to the political feasibility of infrastructure projects and to the equity assessment that should accompany any public investment decision.

2.2.2 Global Evidence: Meta-Analyses

Due to the extensiveness of the literature on transit and property — comprising hundreds of studies across multiple continents — several studies have conducted meta-analyses in an attempted to synthesize this evidence.

For example, Debrezion et al. (2007) analyzed 57 studies to conclude that railway stations have a significant positive impact on property values, with effects varying by property type, distance, and rail service characteristics. Residential properties within 400 meters of a station showed premiums of approximately 4.2%, while commercial properties exhibited larger effects. These findings could be considered a good benchmark to offer perspective to the findings of this thesis.

Mohammad et al. (2013) conducted a meta-analysis of 102 observations from 23 studies. They found that rail projects increase residential property values by an average of 2-8%. More importantly they identified significant heterogeneity across studies, with effects depending on urban density, transit mode, and methodological choices.

Higgins and Kanaroglou (2016) reviewed 60 empirical studies specifically focused on light rail transit — the mode most comparable to Montpellier's tramway — and found that while a majority of studies reported positive effects, the magnitude varied considerably depending on local land use policies and the maturity of the transit system. They emphasized that the strongest capitalization effects occur when transit investment is accompanied by supportive zoning and densification policies.

These meta-analyses consistently find positive effects but also substantial variation. Premiums range from negligible to over 25% depending on context, with the strongest effects observed for heavy rail in dense urban areas with high transit ridership. Such pattern is consistent with economic intuition: transit infrastructure expands a city's transportation network, and where it alleviates congestion or connects previously underserved corridors, it enables urban development and enlargement of economic areas that were previously constrained by accessibility limitations.

However, this development should not be taken for granted. The literature emphasizes that transit investment alone is a necessary but may not be a sufficient condition for value creation, as the strongest capitalization effects are observed when infrastructure is accompanied by supportive land use policies, such as transit-oriented development (TOD) zoning, increased density allowances near stations, and coordinated public realm improvements (Higgins & Kanaroglou, 2016). In the absence of such complementary planning, transit investments may generate accessibility gains without translating them into measurable property value uplift. Integrating the real estate value created into VAN-ES calculations could therefore create a virtuous incentive: city planners would be encouraged to coordinate infrastructure investment with thorough urban planning, precisely because doing so would improve the project's socio-economic evaluation.

2.2.3 Evidence from France

French studies on transit capitalization, while less numerous than Anglo-American research, have seem to point out similar patterns and effects. Infrastructure projects can have meaningful impact on nearby property value. For instance, several studies about tramway lines and metro lines showed significant appreciation of the land using difference-in-differences methods.

In their paper, Boucq and Papon (2007) examined tramway effects in Nantes using hedonic methods. They found modest but positive price premiums for properties with good tramway accessibility while highlighting the importance of controlling for neighborhood characteristics that may be correlated with transit provision.

The IAU-IFSTTAR (2011) study the T3 tramway (Tramway des Maréchaux) in Paris used both hedonic regression and qualitative interviews with real estate professionals. Their findings were nuanced: they detected positive effects within 400 meters of stations but found no evidence of anticipation effects before the line opened—suggesting that tramway (as a lighter mode than RER or metro) may capitalize into prices only upon completion.

Industry sources report from the AFEDIM (2021) that tramway arrivals in French cities typically generate property value increases of 10-15% in surrounding areas. In Strasbourg, the extension of tramway lines A and E to Illkirch-Graffenstaden reportedly increased prices by approximately 5%, while the Grand Paris Express announcements have driven 5-9% increases in affected communes before construction completion.

2.2.4 Methodological Considerations

Estimating the causal effect of transit on property values presents several econometric issues that our methodology must address these include:

Selection bias. Transit lines are not randomly located. They are typically routed through areas with existing development potential, strong travel demand, or political support. Because these characteristics may independently affect property values, I will need to extract only the part that is related to the new infrastructure to avoid overlapping in our computation (As I have previously said, CEREMA framework explicitly warns against double-counting). Naive comparisons between areas near and far from transit may thus confound transit effects with pre-existing differences.

Temporal dynamics. Transit effects may manifest at different stages — public inquiries, officialization, construction, and operation — and, as such, may evolve over time as neighborhoods adapt. The literature distinguishes between « anticipation effects » (price increases upon announcement) and « completion effects » (additional increases upon opening). Our panel data spanning 2015-2025 allows us to examine these temporal patterns for Montpellier's Line 5 but it will be important to re-run the econometric analysis in order to have a more definitive analysis of the « completion effect ».

Spatial dependence. Property prices show evidence of spatial autocorrelation as nearby properties tend to have similar values. Standard regression assumes independent observations, potentially biasing estimates and standard errors. To address this issue, I will have to resort on spatial econometric techniques (spatial lag and spatial error models). (Wooldridge, 2010, Chapter 10)

Difference-in-differences (DID). This approach, which I employ, compares price changes in treated areas (in our case near the new transit infrastructure) to price changes in control areas (far from it) before and after the intervention. Under the parallel trends assumption — that treated and control areas would have evolved similarly absent the treatment — DID identifies causal effects while controlling for time-invariant area characteristics and common temporal shocks. The assumption of parallel trends holds as there is already an extensive transportation network in Montpellier and also

as the tramway line 5 is going through various types of urban areas.

2.3. The Value Capture Debate

2.3.1 Conceptual Framework

If transit investments increase nearby property values, a natural question arises: Should public authorities capture some of this value to help finance the infrastructure that created it?

This question has generated substantial policy debate under the land value capture debate (LVC) (Medda, 2012; Suzuki et al., 2015). The economic rationale for value capture rests on several arguments. First, the land value increase represents a gain for property owners even though they did not perform any activity or take on any risk to create it as the gain derives entirely from public investment. Capturing this value would mean returning all or part of the publicly-created benefits to public entities (George, 1879; Arnott & Stiglitz, 1979). Second, value capture can help finance infrastructure that would otherwise not be financially viable despite positive net social benefits (Peterson, 2009). Third, by taxing land value increments rather than productive activity, value capture may be less distortionary than alternative revenue sources.

Taxing land value could reduce speculation on land appreciation, thus resulting in more realistic valuation of real estate (Oates & Schwab, 2009), while incentivizing property owners to develop or improve their holdings rather than passively benefit from publicly-created appreciation and betting on public investment to improve the neighborhood. However, implementation is, in practice, far more challenging: measuring attributable value increases with precision is difficult, targeted land taxation could discourage beneficial development near transit, and political resistance from affected property owners is to be expected (Smolka, 2013). There is also a longstanding theoretical debate about whether property value increases represent "new" value or simply capitalize benefits already counted elsewhere in cost-benefit analysis. This question first raised by Mohring (1961) is still unresolved in the applied literature (Wardman, 2022).

2.3.2 Global Mechanisms

Capturing value from real estate appreciation can be achieved through many different instruments, each resting on specific assumptions about how value should be measured and who should bear the cost. In practice, there is no single strategy across jurisdictions.

Tax Increment Financing (TIF), common in the United States, captures the incremental property tax revenue attributable to redevelopment within a designated area. This increment finances the infrastructure that catalyzed the development. The Kansas City streetcar, for example, was partly funded through TIF mechanisms that captured the \$2 billion in economic development the line generated (Federal Transit Administration, n.d.).

Special Assessment Districts levy additional charges on property owners within defined areas near transit to fund infrastructure. In Washington DC, the NoMa–Gallaudet U station was successfully funded through this mechanism (OECD, 2022).

Developer Contributions require developers to participate in financing infrastructure improvements that benefit their projects. London’s Crossrail was partially funded through a Business Rate Supplement on non-domestic properties and through Section 106 developer agreements (ITF-OECD, 2024).

Joint Development involves transit agencies partnering with private developers to create mixed-use projects at station locations and share revenue. Hong Kong’s MTR Corporation has used this “Rail plus Property” model extensively, generating sufficient revenue to fund rail construction without operating subsidies (OECD, 2022). Betterment Levies are one-time charges on property owners proportional to estimated value increase. Colombia has used its “contribución de valorización” for decades to fund urban infrastructure (Medda, 2012).

2.3.3 French Context

France is yet to adopt comprehensive value capture mechanisms for transit, that been said, several instruments already in place capture elements of infrastructure-induced land value:

The **taxe d’aménagement** applies to new construction and can be modulated by zone, potentially capturing some development value near new infrastructure. However, it is not specifically linked to transit accessibility improvements. Also, **participations from developers** can be negotiated for contributions to local infrastructure, though these are typically tied to specific developments rather than systematic value capture.

That been said, the absence of explicit value capture mechanisms in France means that property value increases from transit **investment accrue primarily to private landowners**. This has implications for our analysis: as including property value increase in VAN-ES computations could be the start of a

broader debate on the distribution of economic benefits from public investment. Should value capture be implemented, municipalities would then be able to pursue projects that would otherwise fail the VAN-ES test. Even without formal capture mechanisms, **some benefits flow back through increased property tax** (taxe foncière) revenues as assessed values rise and this would be worth estimating.

2.3.4 Implications for VAN-ES

The value capture literature is relevant to our question in several ways. First, it confirms that **transit-induced property value increases are economically significant**, significant enough that multiple jurisdictions worldwide have developed mechanisms to capture them. Second, it raises the question of whether these increases should appear in cost-benefit calculations at all, or **whether they merely reflect capitalization of benefits already counted elsewhere**, such as travel time savings. The consensus view in welfare economics is that property value increases in a competitive land market should approximately equal the present value of accessibility benefits, thus representing the same benefits measured differently rather than additional benefits (Mohring, 1961; Jara-Díaz, 1986). However, this capitalization equivalence holds only under stringent assumptions, such as perfectly competitive land markets, no externalities, and benefits that accrue entirely to property occupants (Kanemoto, 1980).

When these conditions are not met — as is typically the case in real urban environment — a wedge emerges between the user benefits captured in standard VAN-ES calculations and the property value effects observable in the market. In practice, transit creates benefits beyond pure accessibility that standard cost-benefit frameworks struggle to monetize. These include neighborhood amenity effects arising from streetscape improvements and pedestrianization, increased attractivity for specific demographics, option value for households who may use the service in the future, and agglomeration economies generated by the increased density that transit enables (Venables, 2007; Graham, 2007). Furthermore, **high-quality public transportation, by restructuring urban accessibility patterns, can catalyze broader urban redevelopment that would not otherwise occur**. Transit-oriented development projects, mixed-use densification, and neighborhood revitalization often accompany — and depend upon — new transit infrastructure (Cervero & Kang, 2011). The benefits generated by these induced developments are distinct from the direct accessibility gains to existing users, yet they remain unaccounted for in current VAN-ES methodology.

This omission may systematically bias municipal decision-making — and the public during the enquêtes publiques — against comprehensive planning approaches that coordinate infrastructure investment with urban redevelopment objectives as they may not see the real benefits from redeveloping an

entire area. As Banister and Thurstain-Goodwin (2011) argued, the failure to account for wider land use impacts in transport appraisal leads to a structural undervaluation of projects whose primary benefit lies in urban transformation rather than travel time savings. Finally, from a public finance perspective, property value increases carry implications that differ fundamentally from user benefits, regardless of whether they represent "new" welfare gains. **Rising property values expand the municipal tax base, generating additional fiscal capacity** that can be quantified through the *taxe foncière* and the *taxe d'habitation* on secondary residences. They redistribute wealth among property owners, with distributional consequences that merit consideration — particularly when, as our results will show, the beneficiaries include property owners (that are not necessarily living in their properties) in *quartiers prioritaires* who bear none of the investment cost. And they can, in principle, be captured through the various mechanisms described above to help finance the infrastructure that created them (Peterson, 2009).

These fiscal and distributional dimensions are relevant to municipal decision-makers even if one accepts the theoretical argument that property appreciation merely capitalizes benefits counted elsewhere. This paper therefore supports the inclusion of property value effects in an extended VAN-ES framework — not as a replacement for user benefit calculations, but as a complementary measure that captures effects relevant to municipal decision-makers and that may foster investing more towards benefits not fully monetized in standard approaches.

3. Data and Methodology

3.1 Research Design

This part of the paper is about **implementing a difference-in-differences (DiD) framework** to estimate the causal effect of Montpellier’s tramway Line 5 and Bustram A on surrounding property values. The identification strategy exploits the spatial variation in exposure to the new line, that is, properties **located within 500 meters of a station constitute the treatment group** while those **beyond one kilometer serve as controls**, combined with the temporal variation generated by the project milestones at which each line’s expected accessibility gains became credible to the market. The estimation sample draws on DVF transaction records for the five Montpellier postal codes covering the corridor over 2015-2025, with 2020 excluded given the transaction freeze during the COVID-19 lockdowns.

As discussed in the literature review, the DiD approach is well-suited to this setting for two reasons. First, it controls for time-invariant, unobserved differences between neighborhoods — such as historical amenity levels or building quality — that would bias a naive cross-sectional comparison. Second, it absorbs market-wide trends affecting all properties simultaneously, isolating the differential effect of transport proximity. This yields an estimating equation of the form

$$(Price_{it})/m^2 = \alpha + \delta_i Treatment_i + \sum_{t \neq 2018} \mu_t Year_t + \sum_{t \neq 2018} \gamma_t (Treatment_i \times Year_t) + X_{it}\phi + \epsilon_{it}$$

where $Treatment_i$ equals one for properties within 500 meters of the line, $Post_t$ equals one for transactions occurring after the relevant cutoff, and β_3 — the DiD estimator — captures the causal effect of proximity once the line becomes credible. The vector X_{it} includes the property-level controls, spatial amenity variables, and fixed effects detailed in Section 2.4. The code estimates this specification first without controls, then with the full set of controls and fixed effects, and in a log-price variant for a percentage interpretation; an event-study specification (reference year 2018) traces the year-by-year treatment path and tests the parallel-trends assumption, and inference is checked with clustering at the postal-code and 500 m grid-cell levels and with Conley spatial-HAC standard errors (Bartlett kernel, 0.5–1.0 km).

Rather than dating treatment to each line’s opening, the analysis anchors the main specification to the construction-start break, when physical works began and the alignment became effectively fixed —

April 2019 for Line 5. This is the point at which the project crossed the threshold of irreversibility: the earlier political announcements (2009, 2016) carried no binding commitment and were historically reversed, whereas once works were underway the route could no longer be modified and only the timing of delivery remained uncertain. Forward-looking buyers should therefore begin capitalizing the expected accessibility gain from this date rather than from opening; dating the cutoff to opening would absorb this anticipatory adjustment into the pre-period and bias the estimate toward zero. A second cutoff in 2022 tests whether capitalization intensifies as construction advances. For Bustram A, where the timeline is more compressed, **The DiD instead trace the effect across its successive milestones** — the June 2019 "Métronome" announcement, the January 2022 "Bustram" relaunch, the December 2023 construction start, and the May 2025 opening — **to identify the stage at which the market responds.**

3.2 Data Sources

3.2.1 Property Transaction Data (DVF)

Our primary dataset is the **Demandes de Valeurs Foncières** (DVF), France's comprehensive registry of property transactions maintained by the Direction Générale des Finances Publiques (2025). DVF records every notarized real estate transaction in France, providing transaction price, date, property type, built surface area, number of rooms, terrain surface, geographic coordinates, and postal code. I used the full dataset for the Hérault department (département 34) covering January 2015 through December 2025, totaling approximately 36.7 million transaction records.

For the Line 5 analysis, I restricted to residential transactions (apartments and houses) with valid prices, coordinates, and surface measurements within the five postal codes traversed by the line (34000, 34070, 34080, 34090, 34170). Extreme values are trimmed at €500/m² and €10,000/m² to exclude data entry errors and atypical transactions such as intra-family transfers, and the year 2020 is dropped as a COVID-19 confound (579 transactions remain after outlier trimming, but the sharp market disruption makes this year unsuitable for DiD identification).

This yields a working sample of 57,499 transactions, of which 47,306 enter the DiD analysis after excluding the 500m–1km buffer zone.

For the Bustram A analysis, I apply the same filters to the five postal codes along the existing line and its planned extension (34000, 34160, 34170, 34740, 34920), yielding a working sample of 17,390 (Phase 1) and 7,873 (Phase 2) transactions.

Output 1 presents sample size of the dataset used.

Postal codes in the dataset:

```
Code Postal: 34980, Area: 7.37 km2
Code Postal: 34880, Area: 7.03 km2
Code Postal: 34090, Area: 12.94 km2
Code Postal: 34000, Area: 24.11 km2
Code Postal: 34750, Area: 32.16 km2
Code Postal: 34670, Area: 11.88 km2
Code Postal: 34070, Area: 12.50 km2
Code Postal: 34080, Area: 7.56 km2
Code Postal: 34160, Area: 80.51 km2
Code Postal: 34740, Area: 8.82 km2
Code Postal: 34920, Area: 6.10 km2
Code Postal: 34830, Area: 10.77 km2
Code Postal: 34470, Area: 8.51 km2
Code Postal: 34170, Area: 11.19 km2
Code Postal: 34730, Area: 8.99 km2
Code Postal: 34790, Area: 16.72 km2
Code Postal: 34990, Area: 11.25 km2
Code Postal: 34680, Area: 9.17 km2
Code Postal: 34430, Area: 13.37 km2
Code Postal: 34660, Area: 41.06 km2
Code Postal: 34570, Area: 33.76 km2
Code Postal: 34690, Area: 32.27 km2
Code Postal: 34970, Area: 32.28 km2
```

Original dataset: 36,742,897 transactions

Filtered to Line 5 postal codes: 158,247 transactions

Filtered to Bustram A Phase 1 postal codes: 23,165 transactions

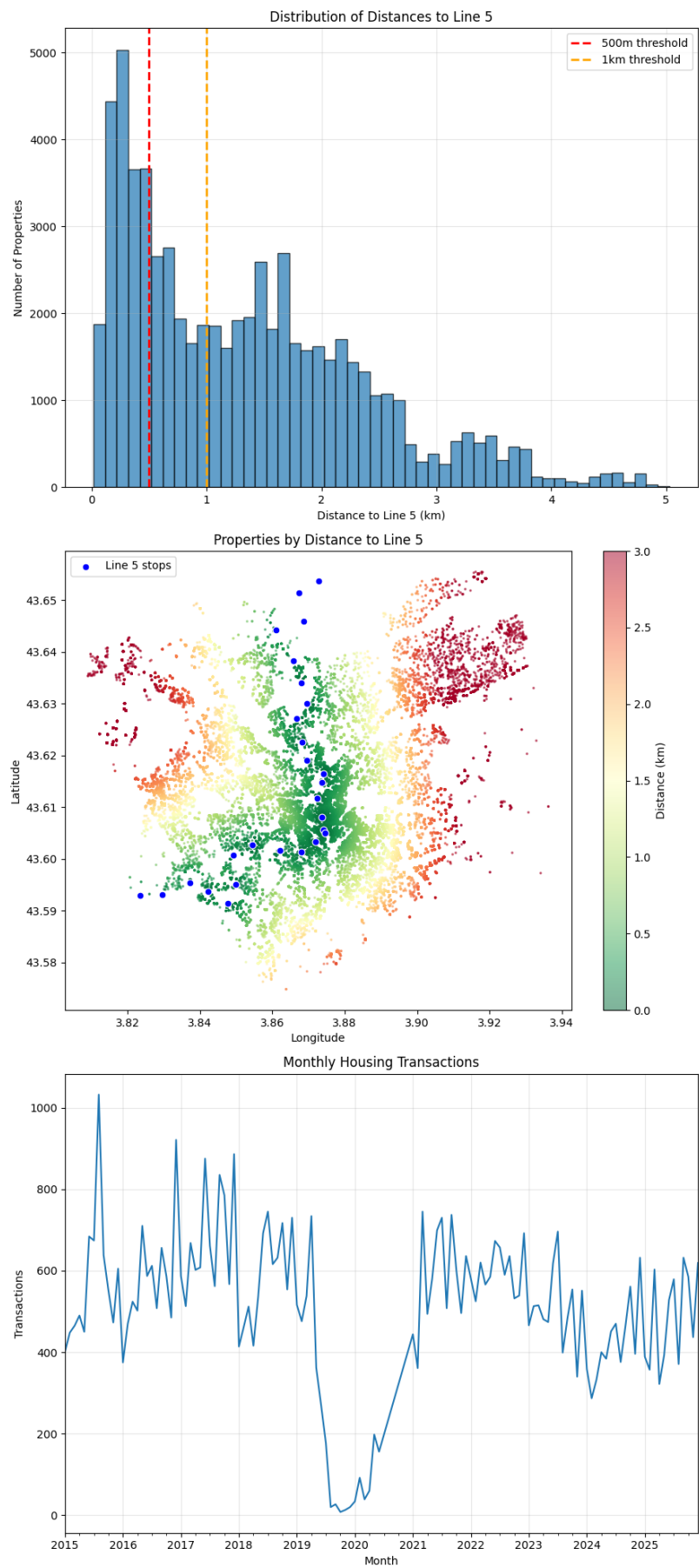
Filtered to Bustram A Phase 2 postal codes: 8,947 transactions

Transactions by year after fix:

```
2015    12815
2016    14955
2017    17989
2018    16330
2019     7215
2020     1135
2021    20914
2022    20933
2023    16776
2024    14238
2025    14947
```

A key limitation of DVF is the absence of detailed property characteristics: floor level, renovation condition, energy performance rating (DPE), parking availability, and building age are not recorded. This is a well-documented constraint affecting all hedonic studies using French transaction data, and it explains why our models achieve R^2 values in the 0.23–0.29 range — comparable to the existing literature using DVF (Fack & Grenet, 2010). This data gap also introduces the risk of conflating transit-induced value appreciation with concurrent improvements in housing quality. For instance, new tram lines often catalyze targeted redevelopment projects where developers construct energy-certified or modernized units to capitalize on the enhanced accessibility. Since DVF does not distinguish between a renovated apartment and an older unit near the tramway, the estimated proximity premium may be upwardly biased, reflecting the combined effect of location and superior building characteristics.

Figure 2. presents summary plot representing DVF dataset and Line 5.



Source: Paul Castet, Code_DiD.

3.2.2 Tramway Line 5 Geographic Data

The precise route of Line 5 is obtained from Montpellier Méditerranée Métropole's open data portal as a GeoJSON file containing 1,200+ georeferenced points along the alignment. This file also includes georeference for all the tramway stops. This is used as a reference for computing which properties are within the 500m range of a tramway station. For each property, my script computes the minimum Euclidean distance to the nearest station, applying a cosine correction for the longitude distortion at Montpellier's latitude (43.6°N, where one degree of longitude corresponds to approximately 80 km rather than 111 km). This is used in `calculate_distance_to_line5` variable.

```
mean_lat = np.radians(property_coords[:, 0].mean())
property_scaled = np.column_stack([
    property_coords[:, 0] * 111.32,          # lat → km
    property_coords[:, 1] * 111.32 * np.cos(mean_lat) # lon → km (cosine
    correction)
])
```

Figure 3. is the scaler used to calculate the distance from each property to Line5 and Bustram A.

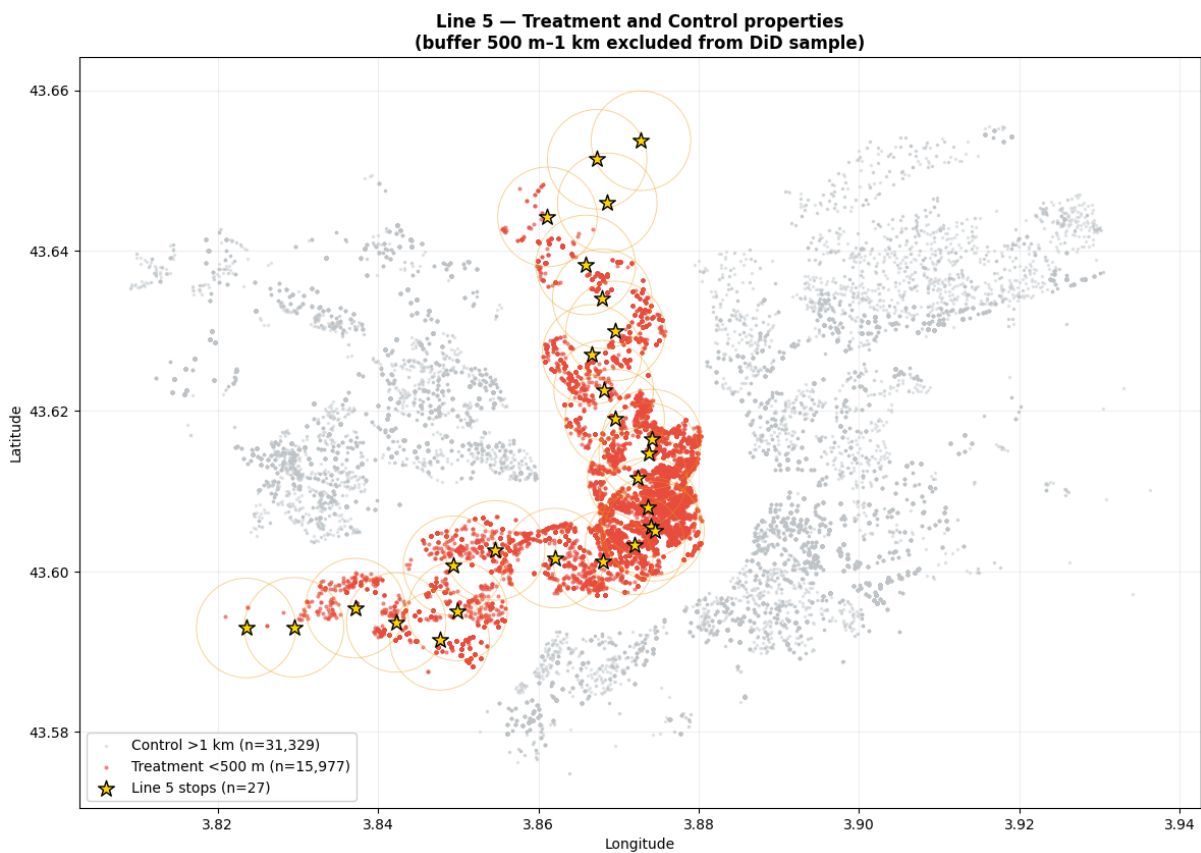
3.3 Treatment and Control Group Definition

The treatment group comprises properties located within 500 meters of a Line 5 station. The control group consists of properties located beyond one kilometer from the nearest station. Properties in the 500-meter to one-kilometer band are excluded to create a clear separation between treated and untreated areas and to reduce the risk of spatial spillover contaminating the control group. This "donut" design is standard in the transit capitalization literature (Gibbons & Machin, 2005; Baum-Snow & Kahn, 2000).

This DiD sample contains 15,977 treated properties and 31,329 controls. **An important feature of the setting is Line 5's complex political trajectory. The project was voted in 2009 and obtained its déclaration d'utilité publique (DUP) in 2013 but was suspended by the incoming mayor in 2014 on the grounds that it was insufficiently financed (to replace the tramway at a fraction of its cost, a bus was proposed). The project was relaunched in 2016 with a substantially modified western alignment (Laverune's mayor team refusing any tramway project passing in the village over fears of densification and changes in demographics), underwent further public consultations through 2017-2018, and finally saw construction begin in April 2019 on the northern section. Construction on the western section followed from October 2021.**

The line was completed in May 2025 and opened to the public on 20 December 2025. DVF data spans 2015–2025, which provides a particularly rich temporal structure for identification. The pre-construction period (2015–2018) captures a phase during which the project’s future was uncertain — suspended, then relaunched with an altered route — and during which I would not expect the market to capitalize benefits that might never materialize. The post-construction period (2019–2025) captures the market’s response once physical works made the project’s completion credible and effectively irreversible. This temporal variation, combined with the spatial variation between treated and control properties, forms the basis of our DiD identification strategy.

Figure 4. Control and Treatment properties map for DiD analysis Line 5 Tramway



Source: Paul Castet, Code_DiD.

3.4 Model Specifications

I estimated four progressively richer specifications to test the robustness of the treatment effect, as follows.

Model 1 (Baseline) includes only the treatment indicator and a linear year trend, capturing the pooled proximity premium across the sample period.

Model 2 (DiD, no controls) adds the post-construction indicator (April 2019) and the treatment $\times$ post interaction — the DiD estimator.

Model 3 (DiD with controls + FE) augments Model 2 with property-level controls (property type indicator, log-built surface), postal code fixed effects, and quarter-of-year fixed effects.

Model 3b (Log specification) uses the same regressors as Model 3 but with $\log(\text{price}/\text{m}^2)$ as the dependent variable, providing a percentage interpretation of the treatment effect.

Model 4 (Late construction DiD) tests whether the effect intensifies during the late construction phase (2022+), when the project's completion became imminent, using the same controls as Model 3.

All models are estimated by ordinary least squares with cluster-robust standard errors at the postal code level (5 clusters). I followed Cameron, Gelbach, and Miller (2008) in acknowledging that clustering with a small number of clusters introduces uncertainty in inference, and cross-validate our results against heteroskedasticity-consistent (HC1) standard errors (White, 1980); the substantive conclusions are unchanged across both approaches.

3.5 Extension to Bustram A: Does Transport Mode Matter?

The Line 5 analysis establishes that tramway infrastructure generates measurable property value effects, capitalized when construction begins. A natural follow-up question is whether this finding generalizes to other transit modes. Montpellier's recently unveiled its “Bustram” network — a Bus à Haut Niveau de Service (BHNS) system — provides an ideal testing ground. By applying the same DiD methodology to the Bustram A line, it is possible to isolate whether the mode of transport itself determines the magnitude of real estate capitalization, holding the metropolitan context and data source constant. The Bustram A project also offers a second, distinct identification opportunity: its planned extension to Castries, announced but not yet under construction, allows us to test whether announcements alone — on an extension of a line that was already inaugurated — can move prices.

3.6 The Bustram A Project

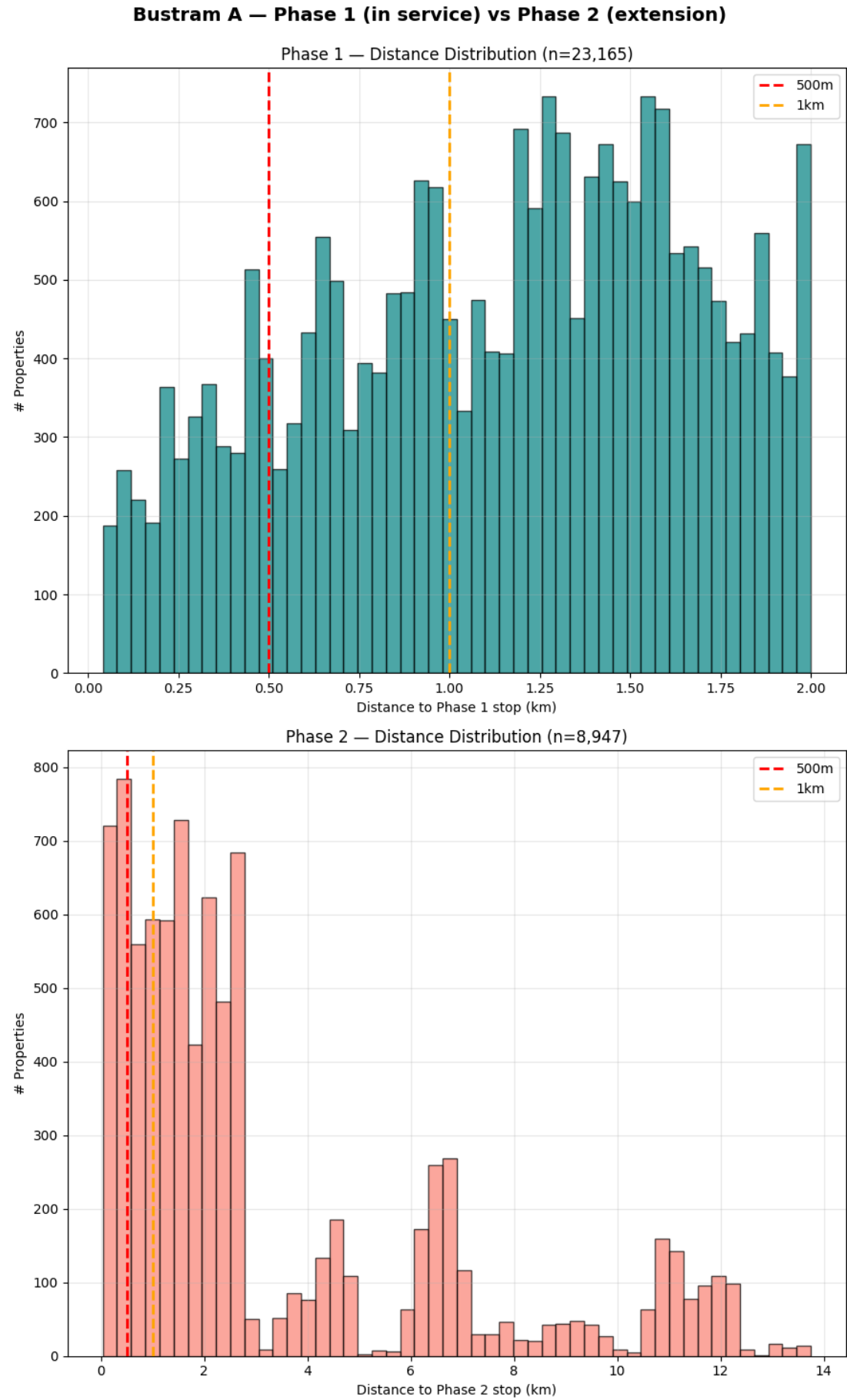
The Bustram project was first announced in June 2019 under the name "Métronomie" as a network of four BHNS lines. But it was only in 2022 that the project was relaunched with a serious commitment under the name "Bustram" with five lines. Section 1 of Line A — the first to be built — runs 4.2 km (of which 2.9 km in dedicated lanes) between the Place de l'Europe intermodal hub in Montpellier and the Notre-Dame de Sablassou park-and-ride in Castelnau-le-Lez, serving 8 stations. The line opened on 23 May 2025 and is operated with 12 electric MAN Lion's City 12E buses at 10 to 15-minute headways. A second section extending the line to Castries (11.8 km total, 20 stations) is planned for September 2028, with preparatory works beginning in November 2025.

The Bustram A corridor presents several important differences from Line 5. First, the infrastructure is lighter: rubber-tired buses operating partially in mixed traffic, compared to a dedicated-track tramway. Second, both terminuses are already served by existing tramway stations (Lines 1 and 4 at Place de l'Europe; Line 2 at Notre-Dame de Sablassou), meaning the marginal accessibility gain from the BHNS may be limited until the line is extended to Castries. Third, the corridor is considerably shorter, yielding a smaller treatment group than Line 5's 27 stations. Fourth the project's two segments are at very different stages of credibility: the existing Section 1 has gone through announcement, construction and opening; the planned extension has only gone through announcement. This naturally separates the analysis into two clearly distinct empirical exercises.

3.7 Data and Methodology

The methodology used to assess the impact of the Bustram A corridor is identical to DiD framework of Line 5, except for the two separate analyses corresponding to the project's two phases.

Figure 5. Summary statistics plot of DVF dataset and BusTram A



Bustram A — Phase 1 (in service) vs Phase 2 (extension)

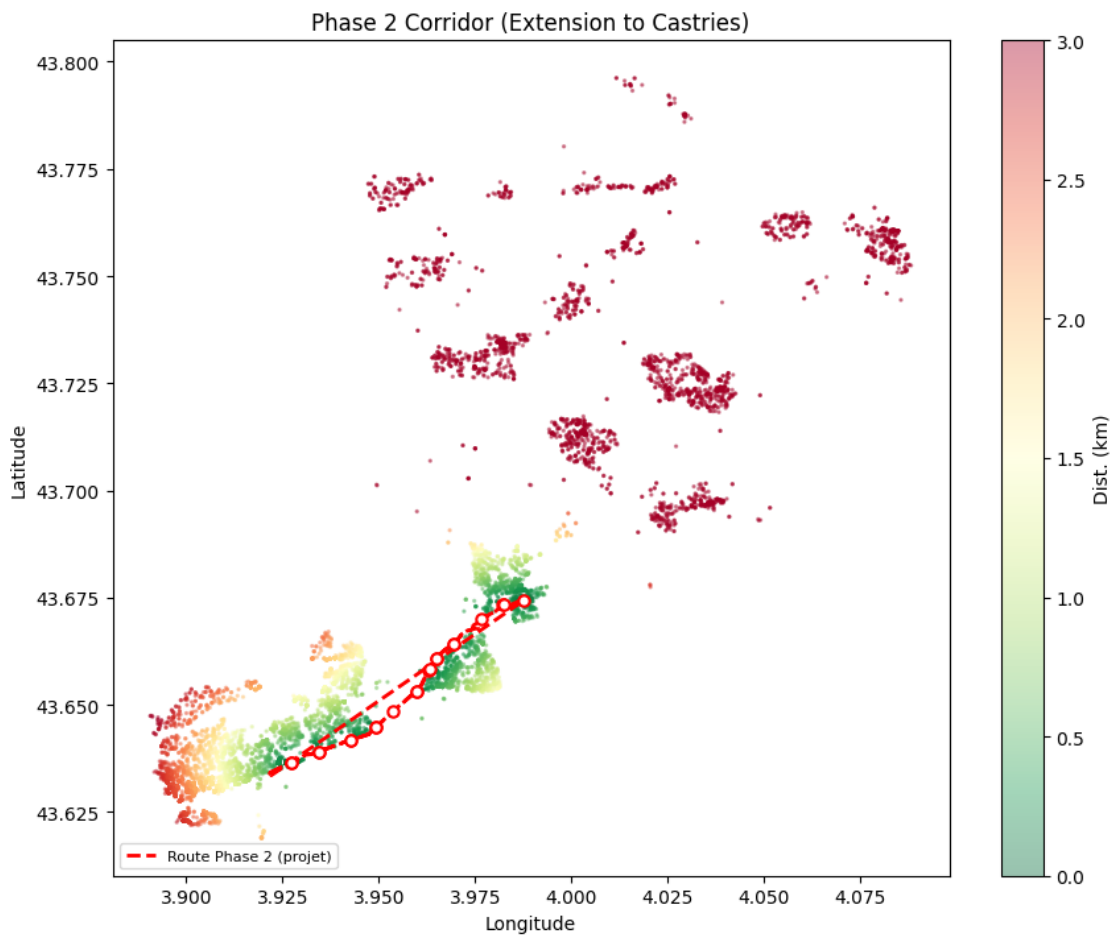
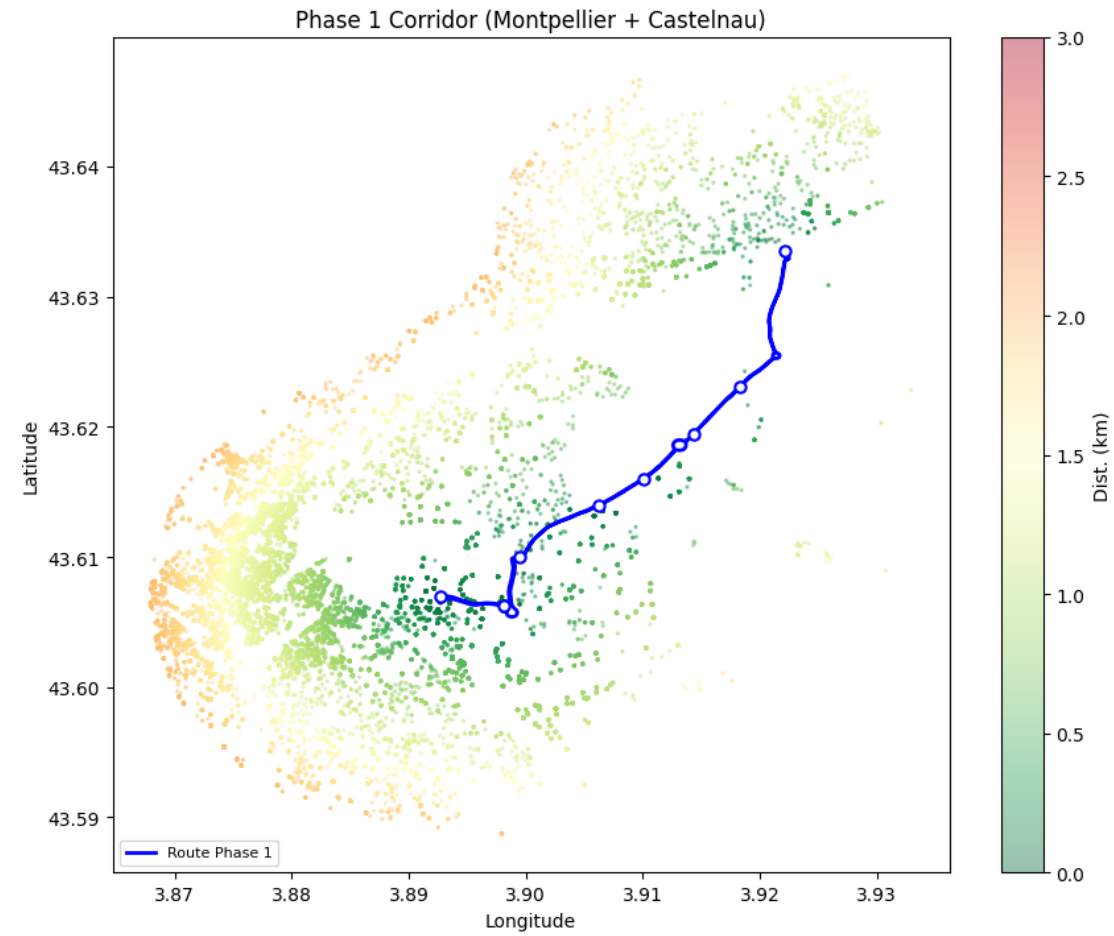
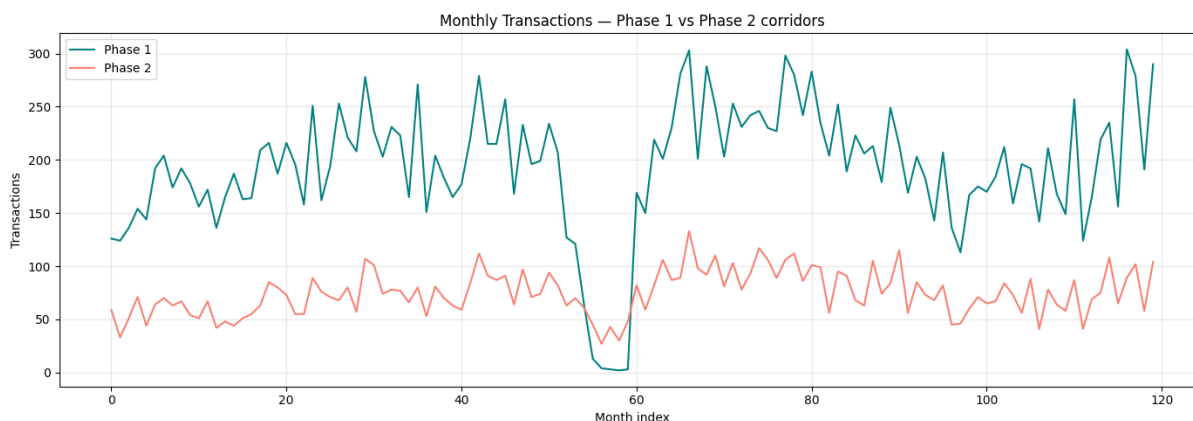


Figure 6. Monthly transactions related to BusTram A DVF sample.

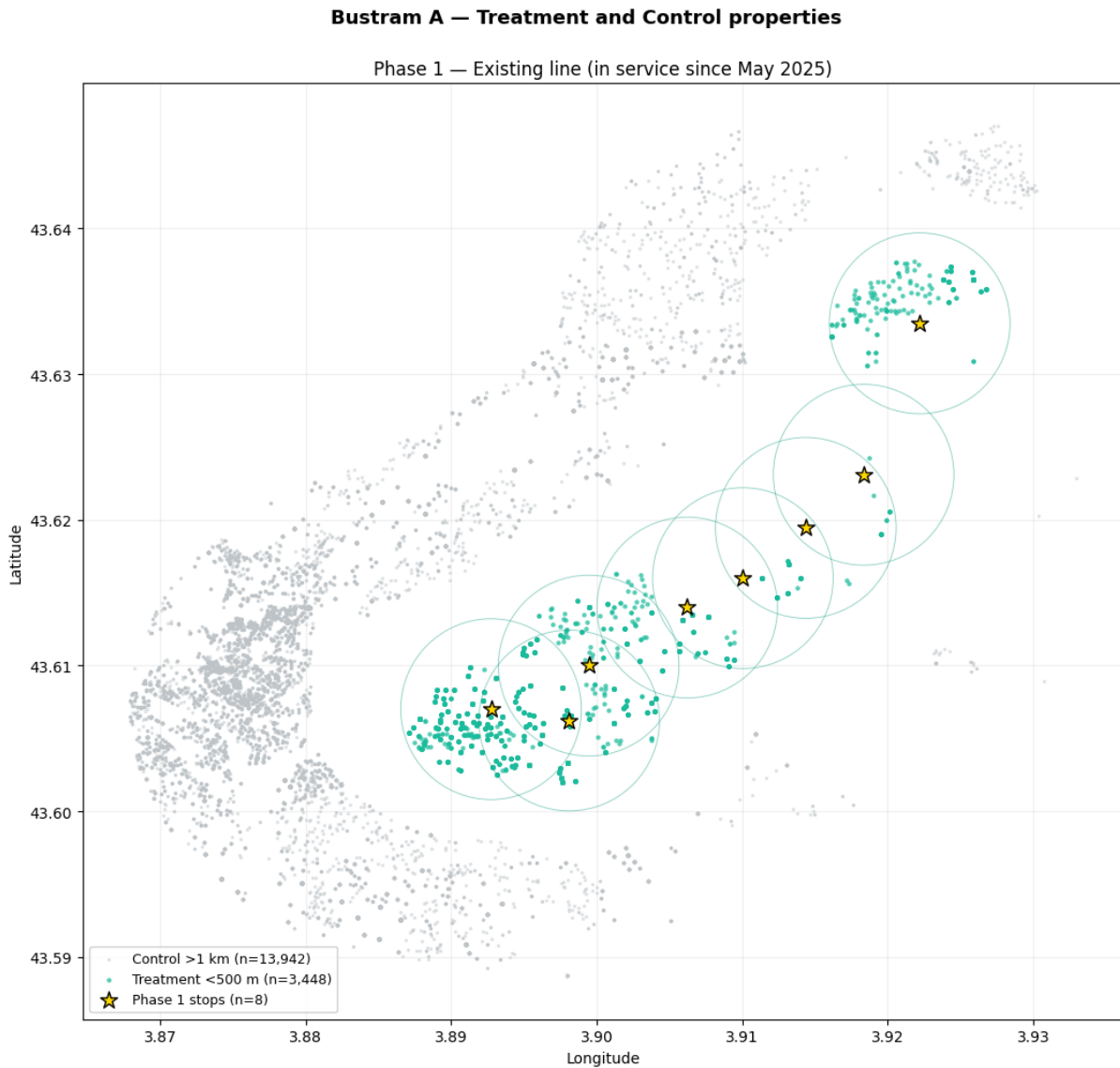


Source: Paul Castet, Code_DiD.

Phase 1 — existing line in service. This analysis uses the 8 stops between Place de l'Europe and Notre-Dame de Sablassou. The relevant postal codes are 34000, 34070, 34080, 34090 and 34170, the same five postal codes used in the Line 5 analysis. The two corridors run roughly parallel, with Line 5 traversing western and northern Montpellier and the Bustram A central section running east from Place de l'Europe through Antigone toward Castelnau-le-Lez. The geometry of the two networks is such that no single property is within 500 m of both a Bustram A Phase 1 stop and a Line 5 stop simultaneously — the closest Line 5 stop (Gambetta–Saint-Denis) sits roughly 1.5 km west of the closest Phase 1 stop (Place de l'Europe), so the 500 m treatment radius do not overlap. Also it is worth mentioning that Phase 1 only runs through Montpellier Metropolitan area and as such the impact of transit infrastructure on real estate prices is already quite high as most properties are already well connected into TaM network.

However, properties classified as Bustram A controls (>1 km from any Phase 1 stop) frequently fall within Line 5's 500 m treatment radius: of the 23,165 properties in the broader Phase 1 corridor, 6,745 (29%) are within 500 m of a Line 5 stop. Without correction, the Bustram A control group would absorb the Line 5 capitalization documented in Section 4.2, biasing the Bustram coefficient downward. All Phase 1 specifications from M3 onwards include the distance to the nearest Line 5 stop and a binary <500 m Line-5 indicator as covariates, absorbing the Line 5 capitalization effect (29 % of Phase 1 corridor properties are also <500 m from Line 5). After applying the price filter (€500-10,000/m²), dropping 2020, restricting the corridor to ≤ 2 km from any Phase 1 stop, and excluding the 0.5-1 km donut buffer, the Phase 1 working sample contains 17,390 transactions: 3,448 treated (<500 m) and 13,942 controls (>1 km).

Figure 7. Control and Treatment properties map for DiD analysis Phase 1 Bustram A



Source: Paul Castet, Code_DiD.

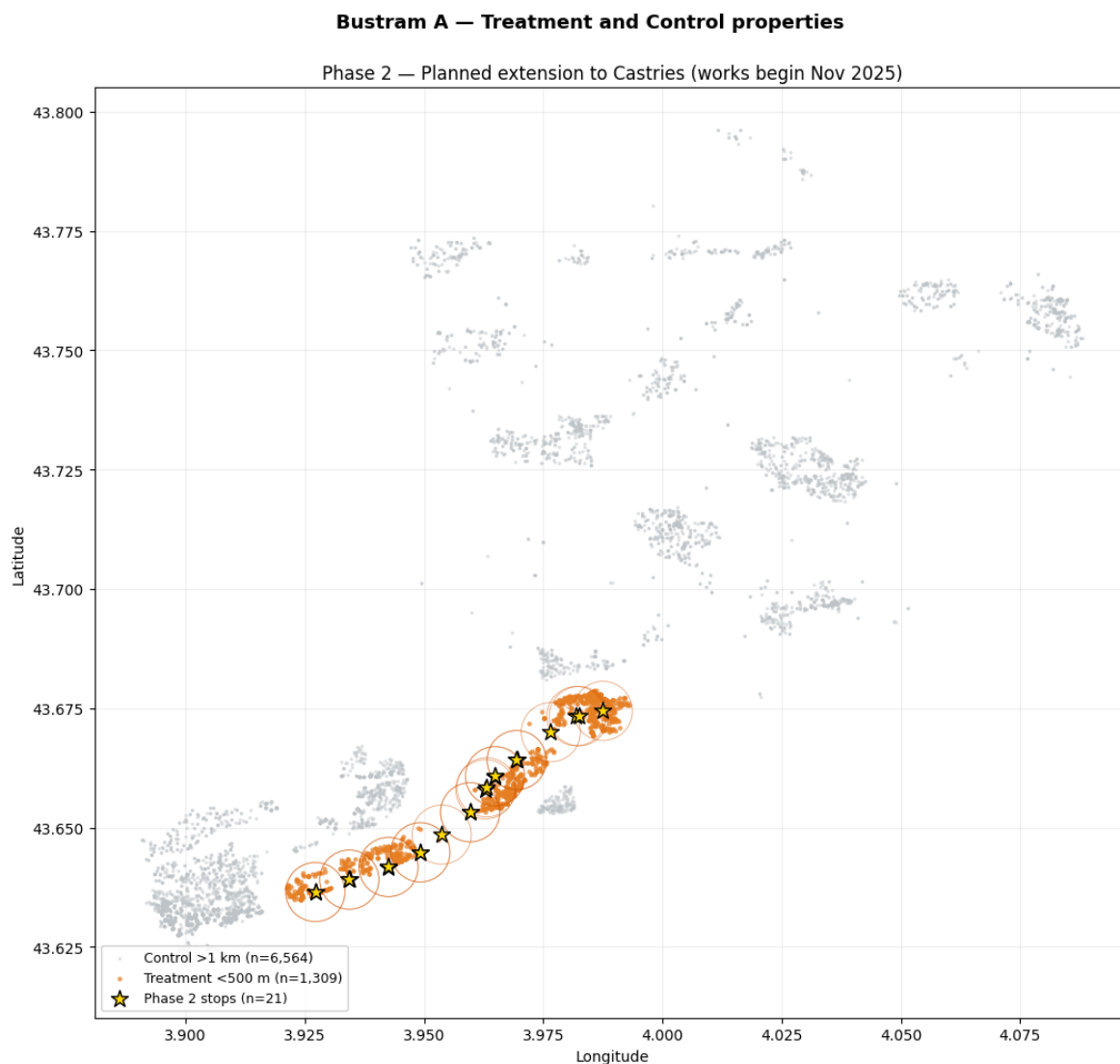
Phase 2 — planned extension. This analysis uses the 21 stops planned between Castelnau-le-Lez and Castries. The relevant postal codes are 34160, 34170, 34740 and 34920 thus covering Castelnau-le-Lez, Castries, Le Crès and Vendargues. Critically, the Phase 2 corridor is geographically distinct from Line 5: zero properties in the Phase 2 sample fall within 500 m of a Line 5 stop. No additional Line 5 controls are therefore needed. The Phase 2 working sample contains 7,873 transactions (1,309 treated, 6,564 controls).

The temporal structure differs between the two phases. For Phase 1, four candidate cutoffs are tested: the June 2019 announcement, the January 2022 Bustram relaunch, the late-2023 construction start, and the May 2025 opening.

For Phase 2, the only meaningful cutoff is the June 2019 announcement, since works on the extension only began in November 2025 leaving fewer than two months of post-treatment DVF data and rendering a works-cutoff specification a placebo by construction.

Phase 2's corridor runs through four communes (Castelnau-le-Lez, Le Crès, Vendargues, Castries) with substantially different pre-2019 price levels and appreciation trajectories. Compositional differences between treatment and control properties are larger than for Phase 1, and the regression results in §4.6 — which absorb postal-code fixed effects, property characteristics, and seasonal controls — will show that the controlled and naïve estimates differ meaningfully in this corridor. It is also worth noting that controls are defined using postal codes (This is relevant since controls groups should only include communes that are part of the Montpellier Metropole.).

Figure 8. Control and Treatment properties map for DiD analysis Phase 2 Bustram A



Source: Paul Castet, Code_DiD.

4. Empirical Results

4.1 Line 5: Descriptive Evidence

Before turning to regression analysis, it is best to review descriptive evidence on the treatment and control groups.

Figure 9. Mean property characteristics by treatment status and period, using 2019 as temporal cutoff.

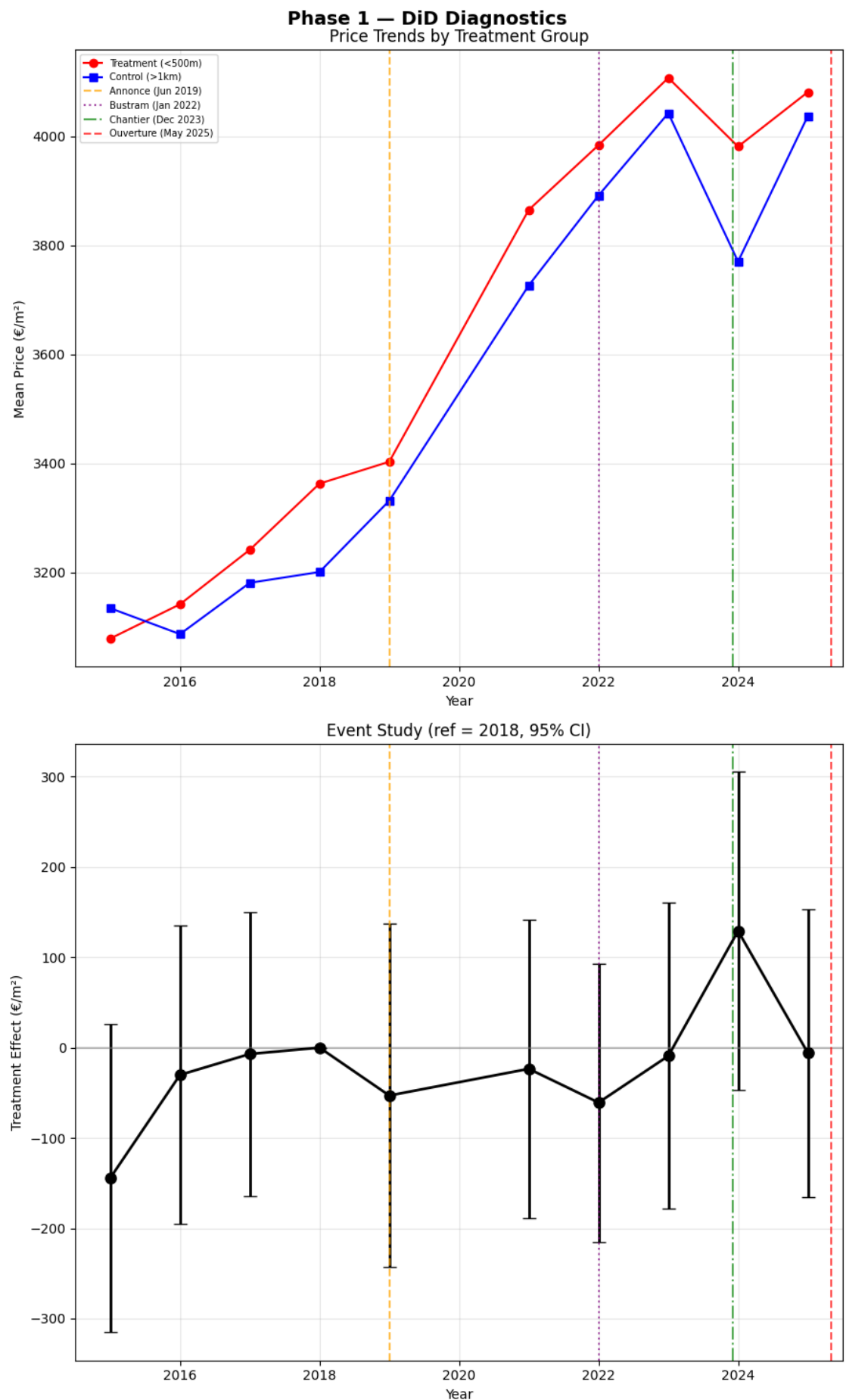
	Treatment (<500m from station)	Control (>1km from station)
Pre-construction (2015–2018)		
Mean price/m ²	€3,241	€3,180
N transactions	7,690	15,170
Mean surface (m ²)	57.9	62.1
Post-construction (2019–2025)		
Mean price/m ²	€3,802	€3,499
N transactions	8,287	16,159
Mean surface (m ²)	58.0	63.5

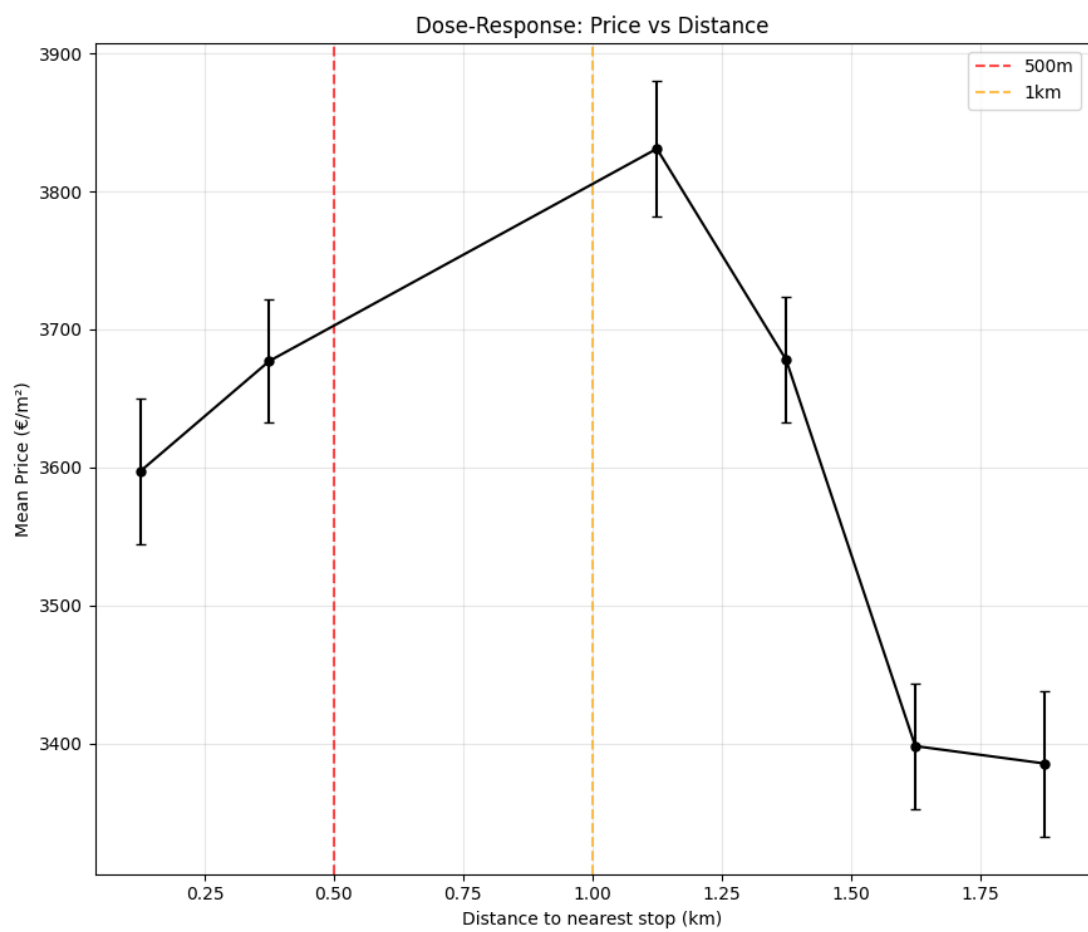
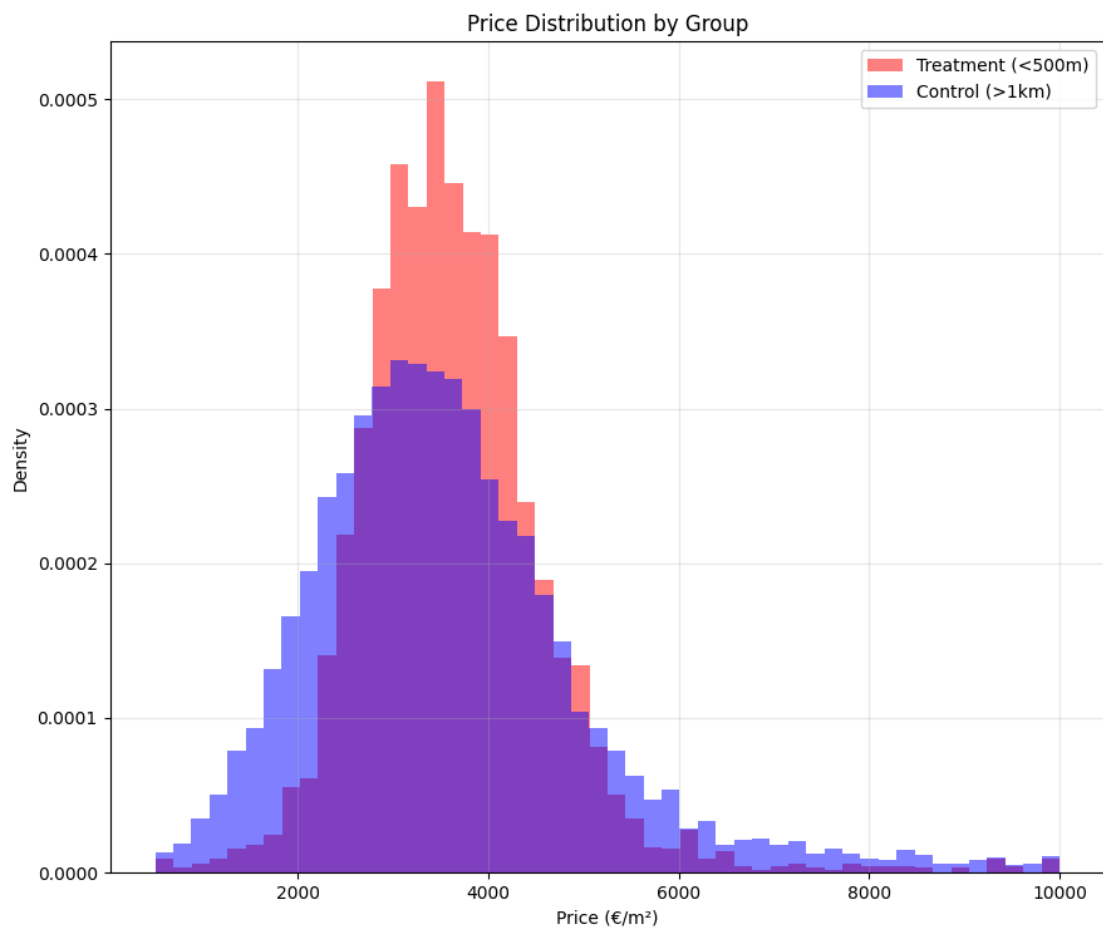
Source: Paul Castet, Code_DiD.

Two patterns emerge from this table. First, treated and control properties had very similar mean prices before 2019 (€3,241 vs €3,180, a difference of only €61/m²) suggesting that proximity to future Line 5 stations did not command a substantial premium before construction began. Second, a clear divergence appeared after 2019: treated properties appreciated by €561/m² while controls appreciated by €319/m², yielding a naïve DiD estimate of approximately €242/m². This raw differential, while not controlling for compositional changes, gives us a glimpse of the regression results.

The near-equality of pre-construction prices is interesting as it implies that the neighborhoods traversed by Line 5 — including lower-income areas such as Estanove, Chamberte, and the Pas-du-Loup — were not systematically more or less expensive than control areas before the project became credible. In other words, the premium emerged specifically with construction, not before. Again, this makes sense because the project had been announced and canceled before.

Figure 10. Event Study Plots for Line 5 DiD





Source: Paul Castet, Code_DiD.

4.2 Line 5: Difference-in-Differences Results

Figure 11. DiD estimates across five specifications.

Specification	Treatment Effect (β_3)	Std. Error	p-value	R ²
M1: Proximity premium only	€185***	13.2	<0.001	0.026
M2: Construction DiD (no controls)	€239***	26.3	<0.001	0.029
M3: Construction DiD (controls + FE, HC1)	€174***	23.8	<0.001	0.236
M3-cl: Same as M3 (clustered SE)	€174***	43.2	<0.001	0.236
M3b: Construction DiD (log, $\times 100=\%$)	+4.58%***	0.69%	<0.001	0.287
M4: Late construction DiD 2022+	€132***	24.3	<0.001	0.235

HC1 robust standard errors (headline); cluster-robust SE (5 postal code clusters) reported separately as M3-cl. Treatment: <500m from nearest Line 5 station. Control: >1km. Buffer zone (500m–1km) excluded. $N = 47,306$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The central finding is that the construction-credibility effect (β_3) is large, positive, and highly significant in Models 2 and 3. The preferred specification (M3) estimates that properties within 500 meters of a Line 5 station appreciated by €174/m² more than controls once construction began in 2019, corresponding to approximately 4.58% of the mean price (M3b, $p < 0.001$). This effect is robust to the inclusion of property-type controls, postal code fixed effects, and seasonal indicators. Adding these controls reduces the point estimate modestly (from €239 to €174) while substantially improving precision (standard error falls from €26.3 to €23.8) and explanatory power (R^2 rises from 0.029 to 0.236). As a robustness check, clustering standard errors at the postal code level (M3-cl, 5 clusters) leaves the point estimate unchanged at €174/m² with a slightly wider standard error of €43.2, and the effect remains highly significant ($p < 0.001$).

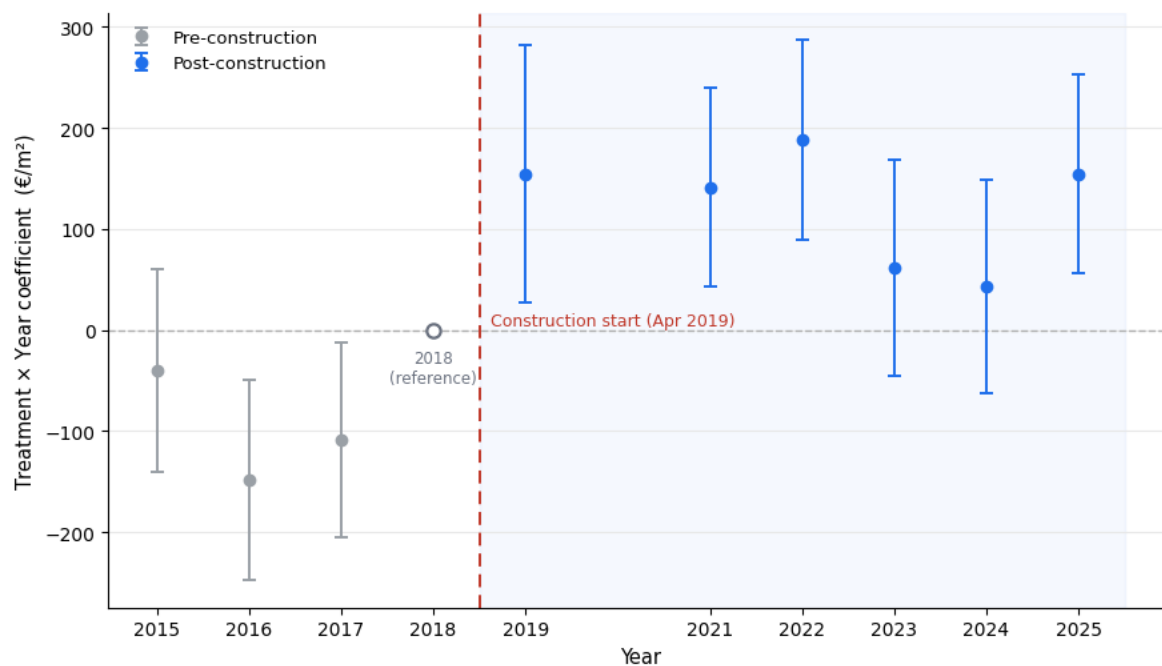
Two further results deserve attention. First, the pooled proximity premium (M1, €185/m², $p < 0.001$) is positive and significant, but understates the true construction-credibility effect. Because M1 pools all years without distinguishing pre- and post-construction periods, it averages a large positive premium (post-2019) with a near-zero or negative differential (pre-2019), producing an estimate — well below the €174/m² identified by M3. The DiD is therefore not revealing an effect that M1 masks both detect a premium — but it correctly isolates its causal timing and produces a sharper, more precise estimate (SE falls from €13.2 in M1 to €23.8 in M3, despite M3 being the richer specification, because the temporal split removes a large source of residual variance).

Second, the Late construction DiD (M4, 2022+, €132/m², $p < 0.001$) is positive and highly significant,

but smaller than M3's €174/m². This ordering is informative: it implies that the bulk of capitalization occurred at the onset of construction in April 2019, with further but diminishing increments as works progressed toward completion. The market did not wait for the tram to run — it priced in expected benefits as soon as physical construction made the project credible and irreversible, then continued repricing gradually as completion risk fell. This front-loaded but reinforcing pattern is consistent with two complementary mechanisms described earlier: rational incorporation of expected future accessibility gains by owner-occupiers (Capozza & Helsley, 1989), and speculative positioning by investors seeking to capture the infrastructure premium ahead of the market (Smersh et al., 2003). Distinguishing between these two channels is beyond the scope of our data, but both imply that the relevant moment for VAN-ES evaluation is the start of construction, not the opening date.

4.3 Line 5: Timeline of events Study

Figure 12. event study (regression) coefficients with 95% confidence intervals



Source: Paul Castet, Code_DiD.

This reveals a clear pattern in how the market priced Line 5 across the project's contested political history. In the years before construction, properties near future stations traded at a discount relative to controls: the 2016 and 2017 coefficients are significantly negative (-€149, $p = 0.003$; -€109, $p = 0.028$), indicating that the corridor was less valued than comparable control areas. This is consistent with the socioeconomic profile of the western alignment, where several areas — including Estanove, La Chamberte, and Pas-du-Loup — are classified as *quartiers prioritaires de la politique de la ville*, with lower median incomes and thinner, less established housing markets than inner-city or southern

Montpellier.

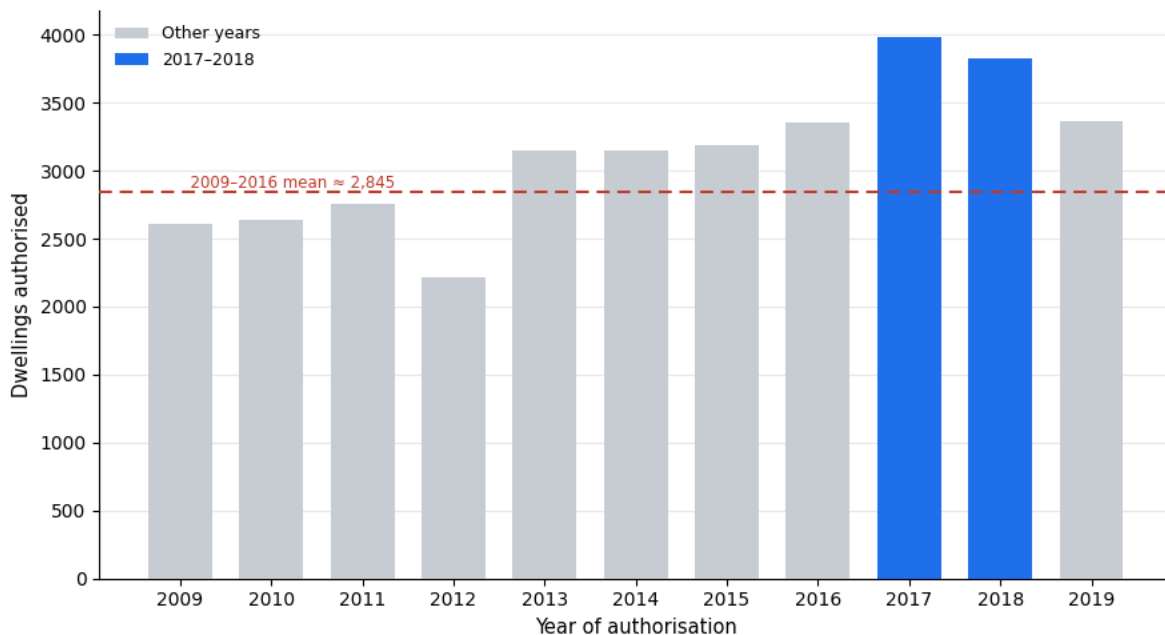
The pattern shifts decisively in 2019. The coefficient moves from -€109 in 2017 to +€154 in 2019, a shift of roughly €263/m² in the treatment-control differential that aligns with the start of physical construction in April 2019 rather than with any earlier political announcement. The premium then deepens: the 2022 coefficient (+€188, $p < 0.001$) is the largest of the construction period, echoing Model 4's evidence of continued capitalization as works advanced, while the imprecise 2023 and 2024 estimates (+€62 and +€43, both insignificant) are consistent with a market consolidating its repricing rather than reversing it. The 2025 coefficient (+€154, $p = 0.002$) is again strongly positive, but it should not be read as an opening effect: Line 5 opened only in December 2025, so the 2025 transaction window is almost entirely pre-opening, and the coefficient can thus be viewed as continued capitalization in anticipation of an imminent, now-visible completion. A genuine opening effect should only be identifiable once the 2026 DVF transactions are released, which had not occurred at the time of writing. Taken together, the event study describes a sustained capitalization dynamic: a front-loaded structural break at construction start, a deepening through the construction period, and a renewed increment as the line approached service.

This sequencing carries a theoretical implication, foreshadowed in the descriptive analysis. The market did not capitalize the project at the 2016 relaunch — indeed, it priced the corridor at a discount through 2016-2017 — and repriced only once construction made the line credible. In a setting where the project had been announced, suspended in 2014, and relaunched with a modified alignment, political commitment alone was insufficient; the costly, hard-to-reverse signal of physical construction was what moved prices. This nuances the standard "announcement effect" documented by Gibbons and Machin (2005) and McMillen and McDonald (2004), whose cases involved projects on comparatively settled political trajectories. Where credibility is in doubt — as for Line 5 between 2014 and 2018 — buyers appear to wait for irreversible investment before pricing in expected accessibility gains.

The pre-period coefficients bear directly on the identifying assumption and warrant a special treatment. The formal parallel-trends test rejects the null of zero pre-2019 interactions ($F = 3.53$, $p = 0.014$), driven by the significant 2016 and 2017 coefficients, making it difficult to claim that parallel trends hold for this specification. Two features qualify how this undercuts the substantive reading. First, the permanent socioeconomic gap between the western-alignment quartiers and the controls — the fact that these are structurally lower-priced areas — is absorbed by the postal-code fixed effects and does not enter the event-study coefficients; what the 2016-2017 estimates capture is therefore a transient deviation, not that standing level difference. Second, the deviation is localized: the 2015

coefficient is small and insignificant ($-\epsilon 40$, $p = 0.434$), so the disturbance is confined to 2016-2017 and coincides with the post-suspension period rather than reflecting a long-run divergence in trends.

Figure 13. Number of building-permit issued between 2019 and 2009



Source: Paul Castet, Code_DiD.

The timing of that deviation coincides with a very active phase of the local housing market, which offers a reading of the pre-period gap. Building-permit records from the Sit@del2 database show that residential authorizations in Montpellier peaked in precisely these years: 2017 was the single highest year for dwellings authorized over the entire 2009-2019 window (3,980 units) and 2018 the second highest (3,825), together running roughly 37% above the 2009-2016 average of about 2,845, with floor area authorized likewise peaking in 2017 (Figure 13). A construction surge of this kind plausibly lifts prices fastest in the better-established, more readily developable parts of the corridor — disproportionately the control areas — while the lower-income western quarters, with thinner markets and less new supply, lag until a credibility shock of their own arrives. On this reading, the negative 2016-2017 differential reflects controls appreciating ahead of treatment in a buoyant supply environment, rather than an anticipatory movement specific to the tramway. This evidence is, however, only suggestive: the permit data are recorded at the commune level and so cannot establish that the additional development was concentrated in control rather than treatment parcels within Montpellier; a city-wide surge affects both groups, and attributing the differential to it would require sub-communal permit locations that the source does not provide.

To assess whether the construction-start effect is an artefact of this anomalous pre-period, I re-

estimated the model on a restricted 2017-2025 window, dropping the 2015-2016 transactions that account for most of the pre-trend violation. The construction-start coefficient attenuates by roughly 18%, from €173.62/m² to €142.94/m² ($p < 0.001$), with the log specification moving similarly (4.58% to 3.57%). This attenuation is consistent with the depressed 2015-2016 baseline having modestly inflated the full-sample differential, which makes the restricted figure the more conservative estimate; the two are not statistically distinguishable, as €142.94 lies well within the full-sample confidence interval of [€127.07, €220.17]. The event-study trajectory is likewise stable across samples — a negative 2017 differential, a sharp break in 2019 (+€158), a peak during advanced construction in 2022 (+€185), and a renewed increment in 2025 (+€146) — indicating that the post-construction dynamic does not hinge on the earliest years.

The restriction does not, however, restore parallel trends. With 2015-2016 removed, 2017 becomes the sole remaining pre-treatment year, and its coefficient stays significantly negative relative to 2018 (-€111, $p = 0.025$), so the pre-trend test continues to reject ($F = 5.06$, $p = 0.025$). The construction-start estimate should therefore be read as robust in sign, significance, and magnitude to the choice of estimation window, while acknowledging that the persistent pre-2019 discount precludes a fully clean causal interpretation on the strength of the event study alone. A formal sensitivity analysis bounding the treatment effect under an assumed magnitude of pre-trend violation rather than treating parallel trends as a binary test, would be the natural next step to quantify how much of the estimated premium survives plausible departures from the assumption.

4.4 the impact of wealth on value effects

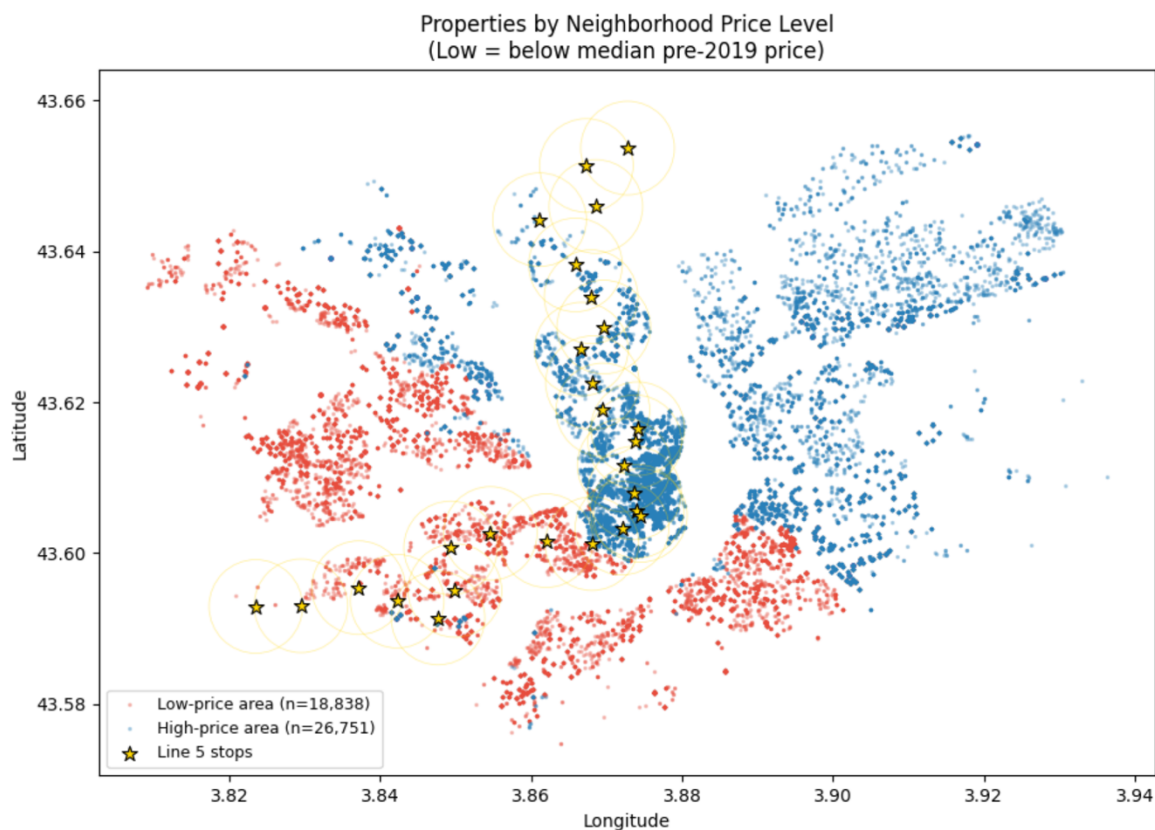
To test whether the construction-credibility effect varies by pre-existing neighborhood wealth, I estimated a heterogeneous DiD that interacts the treatment effect with a binary indicator for below-median postal code price levels. The results reveal a clear distributional gradient. In higher-price areas (postal codes 34000, 34090, and 34170, with pre-2019 median prices at or above €3,146/m²), the DiD effect is €106/m² ($p < 0.001$). In lower-price areas (postal codes 34070 and 34080, with pre-2019 medians of €2,735 and €2,165/m² respectively), the effect carries an additional €104/m² ($p = 0.025$), bringing the total to €210/m², which is roughly twice the premium observed in higher-price areas. The log specification confirms this pattern: a 3.29% premium in higher-price areas, rising to 6.21% in lower-price areas. A split-sample estimation, in which Model 3 is re-estimated separately on each subsample, returns essentially identical figures (€213/m² in low-price areas, €108/m² in high-price areas, both $p < 0.001$), confirming that the heterogeneity is not an artifact of the interacted specification.

These lower-price postal codes encompass precisely the quartiers prioritaires traversed by Line 5's

western section — Estanove, Chamberte, Pas-du-Loup, Ovalie, and Val-de-Croze — where the tramway investment is most transformative to urban connectivity. The finding that these neighborhoods benefited disproportionately is consistent with Kahn (2007), who documented larger transit capitalization effects in poorer station areas, and with the theoretical prediction that marginal accessibility gains are most valued where pre-existing connectivity is weakest. It is also consistent with the front-loaded capitalization pattern documented in our event study: in areas where private alternatives to the new infrastructure are least accessible, the credible signal of an irreversible public investment translates into a sharper repricing.

In summary, this analysis show that real estate premium is not evenly distributed: it is concentrated in areas where property values were lowest, where residents are least likely to own vehicles, where the accessibility improvement was most transformative, and where land redevelopment is most likely to follow. Including a distributional weight in VAN-ES calculations — as recommended by Quinet (2013) for equity-sensitive parameters — would further increase the computed social value of transit investments that traverse disadvantaged neighborhoods, precisely as Line 5 was designed to do. During the discussion section of this paper, I present one interesting model capturing value uplift at the neighborhood level to explore the factors contributing to this phenomenon.

Figure 14. represent Properties by Neighborhood Price level



Source: Paul Castet, Code_DiD.

4.5 Bustram A: Descriptive Evidence

Figure 15. naïve DiD tables for Bustram A Phase 1.

Milestone	Control pre	Control post	Treat. pre	Treat. post	Δ Control	Δ Treat.	Naïve DiD
Announcement (Jun 2019)	3,167	3,887	3,228	3,997	+719	+769	+50
Relaunch (Jan 2022)	3,290	3,942	3,359	4,038	+651	+678	+27
Construction (Dec 2023)	3,476	3,922	3,549	4,037	+446	+488	+42
Opening (May 2025)	3,525	4,073	3,613	4,095	+548	+482	-66

Source: Paul Castet, Code_DiD.

Three patterns are worth noting. First, treated properties trade at a small but consistent level premium over controls in every pre-period, of approximately €60-90/m². This is the reduced-form "proximity premium" that P1-M1 isolates. Second, the naïve DiD is small in every specification. In fact, the treatment-control differential moves by no more than €70/m² across any of the four cutoffs, and changes sign at the opening cutoff. The descriptive evidence thus foreshadows the null announcement, relaunch and opening results reported in §4.6 Third, the naïve construction-cutoff DiD (+€42/m²) is a small fraction of the controlled construction-cutoff DiD (P1-M5: +€101/m²), meaning that compositional differences between treated and control properties (housing type, surface, postal-code fixed effects) materially attenuate the raw difference, and that the regression-based estimate is the more informative one. It is also worth noting that BustramA Phase 1 mostly ran through peri-urban areas (not yet fully developed ones) and therefore the property value increase can also be attributed to the overall redevelopment of the area that came along the bustram A.

4.6 Difference-in-Differences Results

Figure 16. DiD Results for Phase 1

Specification	Treatment Effect	Std. Error	p-value	R ²
P1-M1: Proximity premium only	€87.71***	20.07	<0.001	0.069
P1-M2: Announcement DiD (no controls)	€53.15	40.06	0.185	0.073
P1-M3: Announcement DiD (controls + FE + Line 5)	€48.22	37.69	0.201	0.228
P1-M3b: Announcement DiD (log)	−0.63%	1.06%	0.549	0.273
P1-M4: Relaunch DiD (Jan 2022+)	€54.85	38.16	0.151	0.223
P1-M5: Construction DiD (Dec 2023+)	€100.78**	46.60	0.031	0.230
P1-M6: Opening DiD (May 2025+)	€58.09	68.36	0.395	0.225

Source: Paul Castet, Code_DiD.

*HC1 robust standard errors. Treatment: <500 m from Bustram A station. Control: >1 km. Buffer 500 m–1 km excluded. All DiD specifications from M3 onward control for distance to nearest Line 5 stop and a near-Line 5 indicator. N = 17,390. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Figure 17. DiD Results for Phase 2

Specification	Treatment Effect	Std. Error	p-value	R ²
P2-M1: Proximity premium only	€51.53	37.51	0.170	0.094
P2-M2: Announcement DiD (Jun 2019+)	−€188.60**	77.49	0.015	0.179
P2-M2b: Announcement DiD (log)	−5.22%**	2.10%	0.013	0.191
P2-M3: Works-prep DiD (Nov 2025+) [placebo]	−€269.08	176.83	0.128	0.175

Source: Paul Castet, Code_DiD.

HC1 robust standard errors. Treatment: <500 m from planned Bustram A extension station. Control: >1 km. Buffer excluded. N = 7,873. The works-prep specification is reported as a placebo: it captures only ~2 months of DVF data and the point estimate is uninformative.

Figures 16 and 17 present the DiD estimates for the two phases. The two tables together produce a more nuanced picture than a simple "tramway works, BHNS doesn't" dichotomy would suggest, with the BHNS results varying systematically across project stages and across the two phases. For Phase 1, the announcement (P1-M3), the Bustram relaunch (P1-M4), and the proximity-only specification (P1-M1, once DiD is applied) all produce statistically null DiD estimates. Once construction physically began in late 2023, the DiD estimate becomes positive and significant: €101/m² ($p = 0.031$), or roughly 2.7%

of the post-period mean price. This is materially smaller than Line 5's €174/m² and arrives later in the project lifecycle, but it is not zero — a BHNS construction start does appear to move prices, just less than a tramway construction start does but again it is still worth remembering that the BusTram A was introduced alongside some urban redevelopment near the Eureka pole and, as demonstrated in the literature review, comprehensive urban redevelopment, when combined with public transport infrastructure, is more likely to increase property value than transport infrastructure alone..

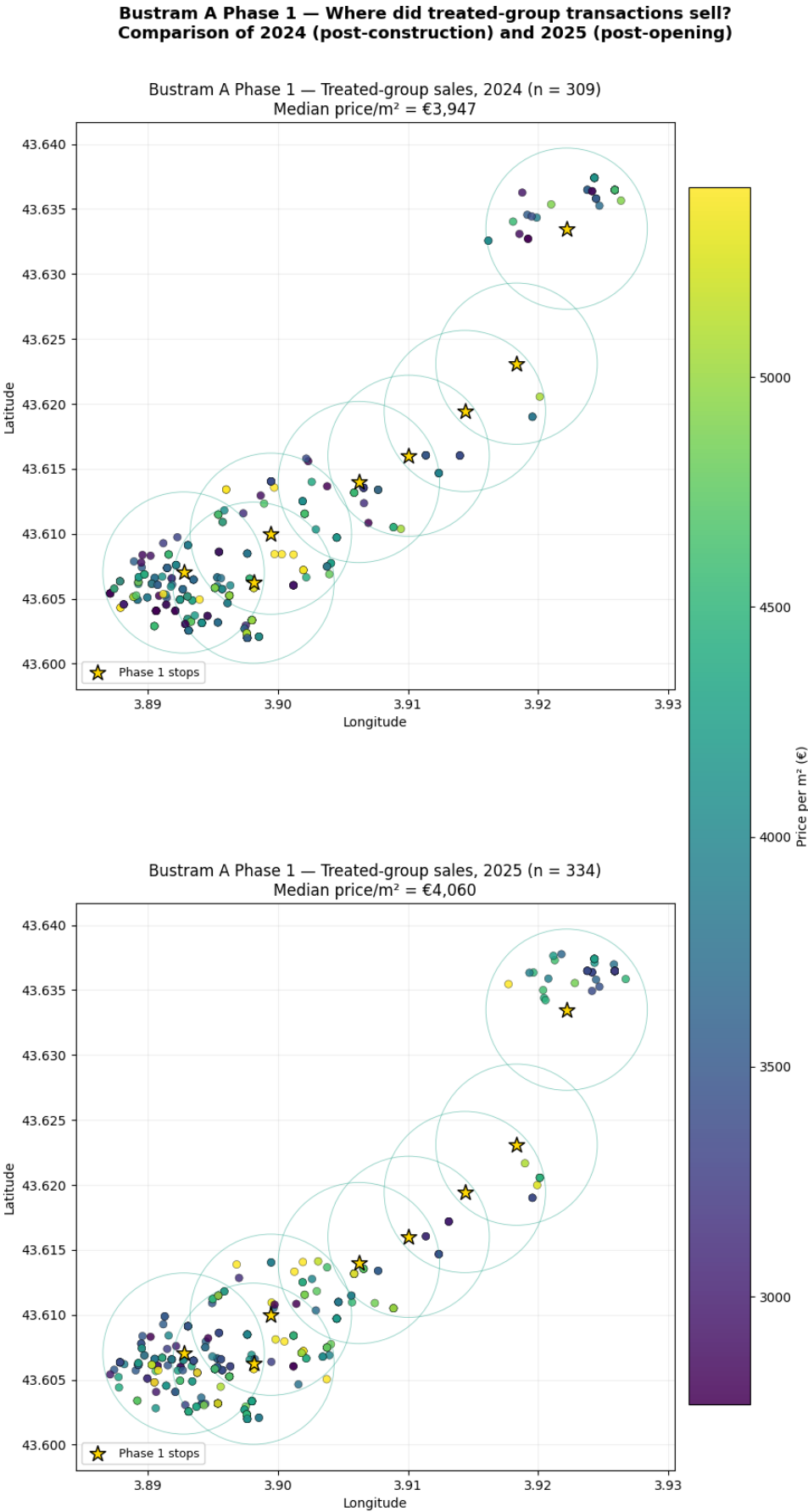
The post-opening specification (P1-M6) returns a smaller and statistically insignificant point estimate (€58/m², SE = €68), but the comparison with P1-M5 should be interpreted with care: P1-M5 and P1-M6 are not nested tests of additional repricing but two separate DiD regressions with different post-treatment windows. P1-M6 isolates the May 2025+ sub-window, which contains only 1,411 transactions and yields a standard error roughly 50% wider than P1-M5's. The smaller P1-M6 coefficient is consistent with the construction-credibility interpretation — that the relevant repricing had already occurred by the time the buses started running — but it is equally consistent with low statistical power, since the post-opening window spans only five months (DVF data for 2026 were not yet available at the time of writing). The two readings cannot be distinguished within this dataset; the DVF 2026 will be needed to settle the question.

For Phase 2, the announcement DiD is negative and significant: properties near the planned extension stations underperformed the rest of the corridor by approximately €189/m² (-5.2% in the log specification) in the years following the June 2019 announcement. The works-prep cutoff (P2-M3) is reported only as a placebo and is uninformative for the data-window reasons noted in §4.6 — the November 2025 cutoff leaves fewer than two months of post-treatment DVF data, and the point estimate is mechanically driven by the small handful of late-2025 transactions in the corridor. Therefore, I believe this result not to be informative.

4.8 Event Study

The +€101/m² estimate from P1-M5 is supported primarily by 2024 transactions (event-study coefficient +€129/m², $p = 0.15$), with 2025 returning a flat year-on-year differential (-€6/m², $p = 0.94$). The spatial distribution of treated transactions is broadly similar across the two years — both cover the corridor evenly from Place de l'Europe to Notre-Dame de Sablassou, with similar price gradients — so the flat 2025 coefficient is not driven by treated transactions sampling different neighborhoods. A more plausible reading is that the treatment-control differential narrowed in 2025 either because metropolitan-wide appreciation accelerated outside the corridor, or because the 2024 coefficient itself was on the high end of the year-by-year noise band.

Figure 18. Color gradient map of real estate sales prices for 2024 and 2025 in the treatment area.



Source: Paul Castet, Code_DiD.

The Phase 2 sample is too small for a fully saturated year-by-year event study to be informative, but the parallel trends assumption can be tested by jointly examining the pre-2019 interaction terms; the test does not reject.

4.9 Comparative Analysis: A Credibility Ladder, Not a Mode Dichotomy

The juxtaposition of Line 5, Bustram A Phase 1, and Bustram A Phase 2 reveals a more granular pattern than a simple mode dichotomy would suggest. Figure 19 summarizes the comparison.

Figure 19. Comparative.

	Line 5 (tramway)	Bustram A — Phase 1 (in service)	Bustram A — Phase 2 (extension)
Mode	Light rail, dedicated track	BHNS, partially dedicated lanes	BHNS, planned
Length	16.3 km	4.2 km	11.8 km
Stations	27	8	21
Status as of end-2025	Opened Dec 2025, full chantier	Opened May 2025, construction visible	Announced 2019, works begin Nov 2025
Credibility signal observed	Construction (Apr 2019)	Construction (Dec 2023)	Announcement only
Headline DiD coefficient	€174/m ² *** (4.58%)	€101/m ² ** (~2.7%)	-€189/m ² ** (-5.2%)
Treatment group	15,977 transactions	3,448 transactions	1,309 transactions
Parallel trends	Marginal rejection at 5%, driven by 2016–2017 level discount	Validated (F = 1.22)	Not rejected

Source: Paul Castet based on data from *Enquête Publique TramL5Montpellier* H – Evaluation socio-économique Tableau 4, 2013 ; Bustram de Montpellier 2026.

Read as a hierarchy of credibility signals, the first two columns of Figure 19 deliver a coherent story. A heavy, costly, irreversible work on a tramway moves prices most, while a lighter but visible BHNS work moves prices by about half as much. This contrast goes beyond the pure "service quality" interpretation would allow. At launch, Line 5 runs at 12-minute headways during the day (TaM, 2025) and Bustram A at 10 to 15-minute headways, making the level of service offered to passengers comparable across the two modes. **Frequency and accessibility thus cannot account for the magnitude difference observed in the data** as both service offer at-level boarding, sufficient capacity and Air-conditioning.

What separates the two infrastructures is what they look like, what they cost, how permanent and

how structural they appear — rails, catenary and dedicated stations on one side; rubber-tired buses on partially shared lanes on the other. The implication, advanced cautiously and inviting further research, is that mode itself carries a perceptual premium that goes beyond the transport service it provides. A tramway and a BHNS can deliver similar accessibility while signaling very different things to the property market about the durability of the public commitment. This is consistent with the broader literature on rail-mode premia (Mohammad et al., 2013) and complements the construction-credibility argument developed in §4.4. Rail capitalizes more than bus not only because rail is harder to remove once installed, but also, plausibly, **because the market reads the act of installing it as a stronger statement of intent to redevelop an area.**

The Phase 2 column is more difficult to interpret. The negative announcement coefficient ($-\text{€}189/\text{m}^2$) is statistically significant at the 5% level ($p = 0.015$), but it is identified off a single shock, in a smaller sample, with no construction-stage estimate to compare against. My reading is that the Phase 2 result is most likely noise, a chance feature of the four-commune extension corridor's heterogeneous appreciation trajectories rather than a genuine market response to the June 2019 announcement. I nonetheless retain the result in the analysis because the sign and timing of the coefficient are at least suggestive of a phenomenon worth investigating once construction-period data become available.

Also, the negative coefficient reflects compositional features of the four-commune extension corridor. Castries, Le Crès, Vendargues and Castelnau-le-Lez follow distinct appreciation trajectories that the controls in P2-M2 may absorb imperfectly, and the small treatment group ($n = 1,309$) leaves the estimate vulnerable to the influence of a handful of unrepresentative transactions. If a residual genuine effect exists beneath this noise, two interpretations would be plausible: NIMBY anticipation in communes where opposition to densification, or a market reading of the announcement as commitment to disruption without credible commitment to delivery. I do not make any claims about the Phase 2 estimates establishing either mechanism as it will only be possible to draw conclusions once a construction cutoff produces a second identifying shock against which the announcement coefficient can be benchmarked.

All in all, three mechanisms help interpret the well-identified Line 5 vs Phase 1 gradient, with a fourth, more speculative observation flagged for completeness regarding Phase 2.

First, **permanence and credibility of investment.** A tramway requires dedicated rails, overhead catenary and purpose-built stations, that is expensive to build infrastructure and effectively irreversible once installed. A BHNS, with rubber-tired buses on lanes that can be repurposed, is a softer commitment, as such the market may rationally — or unconsciously — discount the permanence of

the service. This single mechanism is consistent with both the Line 5 vs Phase 1 magnitude difference and the broader literature on mode-specific capitalization (Mohammad et al., 2013).

Second, **marginal accessibility gains**. The Phase 1 corridor connects two points already served by tramway lines, through an area with existing bus service. The incremental accessibility improvement is modest, which may explain why the Phase 1 works have moved prices but only by about half what Line 5's did. Line 5, by contrast, opened an entirely new corridor connecting western Montpellier to Clapiers immediately, with no equivalent uncertainty about delivery.

Third, **scale and urban transformation**. Line 5's 16.3 km alignment involved substantial streetscape improvements — new bridges over the Lez and the Lironde, demolition and reconstruction at the place de Chine, complete redesign of major intersections — alongside new public spaces, cycling infrastructure and density changes amounting to a €440 million investment in the urban fabric. Phase 1 of Bustram A involved more limited physical transformation, reducing the amenity externalities that compound into property values. In fact most of the urban redevelopment concentrated around the new development area of Eureka alongside Phase 1.

These results are consistent with the international meta-analytic evidence reviewed in §2.2.2. Mohammad et al. (2013) found that rail projects generate larger property value effects than bus-based systems, and Debrezion et al. (2007) documented a clear hierarchy by mode. Within-city comparison confirms this hierarchy empirically and the Line 5 / Phase 1 contrast adds a finer dimension: it is not merely the mode that matters, but the credibility and permanence signal that the mode and the project stage convey to the real estate market. A BHNS under construction beats a tramway that has only been announced, all the more so as the BHNS project was combined with comprehensive urban redevelopment around the Eureka pole. Whether announcements alone can move prices in the opposite direction, as the Phase 2 estimate hints at without demonstrating, is a question this paper leaves for future work with richer post-2025 data and more cities studied.

5. Discussion

This discussion argues that the central case for incorporating real estate appreciation into VAN-ES is not accounting completeness but incentive design: because capitalization is largest where transit is coordinated with redevelopment, a metric blind to capitalization is blind to the value of that coordination, and reforming it would make the socio-economic score reward integrated planning directly

5.1 From accounting to incentive design

The empirical results of this thesis — a construction-credibility premium of roughly €174/m² for tramway Line 5, a smaller and contingent effect for the Bustram A BHNS, and a descriptive gradient analysis linking the size of the capitalization signal to the surrounding urban fabric — converge on a the below argument about how transport investment should be evaluated.

The logic can be divided in six aspects. First, the value transit investment creates in the land market is largest not where a line is built in isolation, but where it is built alongside coordinated redevelopment of the surrounding area — a regularity predicted by the spatial-equilibrium literature (Alonso, 1964; Muth, 1969) and consistent with both the GBM evidence presented below and the redevelopment that accompanied Line 5's western alignment. Second, the French socio-economic evaluation framework (VAN-ES) currently assigns no value to this capitalization. Third, a project's VAN-ES score is therefore not sensible to whether the transit investment is paired with the land-use coordination that would maximize its social return as the metric does not account for the difference. Fourth, incorporating a real estate appreciation term would make the score rise precisely with the degree of coordination, since that is where measured capitalization is largest. Fifth, because VAN-ES gates public funding and political approval, a metric that rewards coordination would, over time, change what gets built — steering decision-makers toward the integrated transit-and-redevelopment projects that the literature identifies as delivering transit's full potential (Higgins & Kanaroglou, 2016). Sixth, omitting real estate appreciation does not prevent it from occurring — the land market reprices regardless — but it does remove the issue from view at the moment it could be deliberated. By leaving capitalization unquantified, the current framework forecloses any productive debate during the enquêtes publiques about how much value a project creates and how that value should be distributed. Made explicit, that value becomes a legitimate object of negotiation — for instance, a basis on which quartiers prioritaires de la politique de la ville might secure additional social housing to offset the displacement pressure that rising land prices would otherwise impose.

This is why the reform proposed here is best understood as an instrument of urban planning rather than of project accounting. The deeper principle is general: metrics shape the decisions they are used to make, therefore effects a metric omits do not thereby become smaller — they simply cease to inform the choices the metric governs. The remainder of this discussion develops the interpretation of the results (Sections 5.2–5.5), consolidates their implications for VAN-ES reform (Section 5.6), and sets out the limitations that bound them (Section 5.7).

One caveat frames the entire argument. The claim that coordination causes larger capitalization is one the present data support by association rather than prove by identification: the GBM is descriptive, and — as Sections 5.4 and 5.7 detail — the Line 5 and Bustram effects cannot be cleanly separated from the redevelopment that accompanied them. The incentive argument does not, however, require a proven causal lever. It requires only that measured capitalization be larger where coordination occurs, which the data show; aligning the metric with that observed pattern improves on a metric that ignores capitalization entirely, even if part of the association reflects the bundling of accessibility and redevelopment rather than a one-way effect. Indeed, if accessibility gains and planning context are jointly productive of land value rather than separable, that is itself the strongest reason to reward their combination rather than transit construction alone.

5.2 Line 5: Interpretation — Construction Credibility is not equal to Announcement

The results admit a clear economic interpretation that departs from the standard anticipation narrative. **The market did not capitalize Line 5's benefits at the political relaunch (2016). It capitalized them when construction began (2019),** making the project's completion credible and effectively irreversible, and continued to price in further gains as works progressed. This construction-credibility interpretation is supported by four pieces of evidence.

First, the DiD coefficient is significant and economically meaningful (€174/m², ~4.58%) using April 2019 as the temporal cutoff. Second, the event study shows a structural break between 2017-2018 (negative or zero coefficients) and 2019+ (positive, significant coefficients), with no comparable break at the 2016 political relaunch. Third, the late-construction DiD (M4, 2022+) is positive and highly significant (€132/m², $p < 0.001$), indicating that capitalization was not a one-time repricing at the onset of construction but followed a progressive accumulation pattern: the decisive break occurs in 2019 when irreversible physical works begin, and the premium deepens through 2022 as the project's completion becomes increasingly certain. This is consistent with the event study, where the 2022 coefficient (+€188, $p < 0.001$) is the largest in the post-treatment period. Taken together, Models 3 and 4 describe a front-loaded but gradually reinforcing capitalization profile — not a single jump, but an increase that

tightens as construction risk goes down. Fourth, the project's political trajectory — with its suspension, relaunch, and route changes — provides a natural explanation for why earlier announcements failed to trigger sustainable capitalization. Political commitments lacking costly, observable follow-through were simply not credible enough for the market to act on.

Even the DUP may no longer carry the credibility it once did as recent infrastructure projects have shown. The A69 motorway between Toulouse and Castres obtained its DUP in July 2018 and began construction in 2023; in February 2025, the tribunal administratif de Toulouse nullified its environmental authorization and ordered the immediate cessation of works, ruling that the project did not meet the threshold of *raison impérative d'intérêt public majeur* (Tribunal administratif de Toulouse, 2025). This was the first judicial cancellation of a French motorway project at such an advanced construction stage. The LGV Bordeaux-Toulouse high-speed rail line obtained its DUP in 2016, yet as of early 2026 its financing remains unresolved, the Conseil d'Orientation des Infrastructures has reopened the question of its viability, and regional authorities have publicly questioned whether the project will ever materialize (France Bleu, 2026). In both cases, the DUP proved insufficient to guarantee completion — a reality that rational real estate markets are likely to internalize, and one that further reinforces the case for identifying the credibility-establishing event empirically rather than by administrative convention.

5.3 Line 5: Implications for VAN-ES

The finding that tramway Line 5 generates a construction-credibility premium of approximately €174/m² (~4.58%) has direct implications for the incorporation of real estate effects into VAN-ES calculations.

To gauge the magnitude of the omission, this premium is scaled to the residential stock within 500 meters of a Line 5 station. The relevant quantity is the standing stock that carries the premium — the dwellings physically located in the buffer — not the flow of transactions observed over the sample. A naïve product of the 15,977 treated transactions, their mean built surface (58.1 m²) and the premium yields roughly €161 million, but this is best read as a lower bound: it counts realized sales rather than dwellings, so it both double-counts units that sold more than once and, more consequentially, omits the far larger stock that never transacted over 2015–2025. Because the standing stock is not recorded in DVF, it is estimated two independent ways that rest on different assumptions and therefore bracket the true value. The first infers the stock from transaction volume alone: at a residential turnover rate of approximately 3.5% per year (IGEDD 2026), the 15,977 sales observed over ten years imply a standing stock of $15,977 \div (0.035 \times 10) \approx 45,600$ dwellings, which at 58.1 m² and €174/m² gives an

aggregate uplift near €460 million. The second uses geometry alone and makes no turnover assumption: the union of the 27 stations' 500 m buffers covers 12.1 km², which at the Montpellier commune-average density of roughly 2,900 dwellings/km² implies some 35,200 dwellings and an uplift near €355 million. The two estimates bracket the true figure from opposite directions — the turnover route likely overstates the standing stock, because the treated sales include first-sales of newly-built units that are net additions rather than churn, so the effective turnover exceeds the assumed 3.2%; the density route likely understates it, because a transit corridor is denser than the commune average the calculation applies. Together they place the residential real estate uplift attributable to Line 5 station proximity on the order of €355–460 million, roughly 80–105% of the line's €440 million construction cost. Getting a more precise figure would require the Fichier Fonciers from CEREMA that is not publicly available.

Figure 21. STOCK-BASED AGGREGATE UPLIFT

Treated transactions: 15,977 mean surface: 58.1 m ² observed years: 10				
Premium (M3): €174/m ² [95% CI 127, 220]				
[1] turnover	:	45,649 dwellings	→ € 460 M	(leans high)
[2] area×density	:	35,194 dwellings	→ € 355 M	(12.1 km ² × 2,900/km ² ; leans low)
=====				
STOCK-BASED BRACKET : €355–460 M (81–105% of €440 M cost)				
LOWER BOUND (txn) : €161 M (37% of cost)				
=====				
Uplift (€M) by premium:				
		CI-lo €127	point €174	CI-hi €220
turnover	337	460	584	
area×density	260	355	450	
Turnover τ ∈ [3%, 4%] → stock 39,942–53,257, uplift €403–537 M				
Source: Paul Castet, Code_DiD.				

This range represents only residential properties within 500 meters of the 27 stations in the five postal codes studied and is therefore conservative. The effect extends — with attenuation — beyond 500 meters, applies to commercial properties, and covers parts of the existing network outside our five postal codes that benefit from the addition of a new line (subsequent studies may be needed to quantify this spillover). The sample also ends before the line's inauguration in December 2025, so any opening effect is excluded. A direct count of residential premises within the buffer — available from the Fichiers Fonciers (CEREMA) but not from DVF — would replace the two estimates above with a register figure and is the natural refinement (Section 5.7).

Two qualifications govern how this figure should enter project evaluation. First, it is a scale marker, not a benefit to be added to the VAN-ES directly. Under standard capitalization equivalence (Mohring, 1961), a large share of it re-expresses the accessibility gains the framework already records as travel-

time savings; adding the full amount would double-count. What is genuinely additional — and what motivates inclusion — is the wedge capitalization captures beyond pure accessibility: streetscape amenity, agglomeration, option value, and the induced redevelopment documented in Section 5.5, none of which standard VAN-ES monetizes. Second, even setting the welfare-accounting debate aside, neither the capitalization nor the additional tax foncière revenue the uplift would generate appears anywhere in the VAN-ES calculation for Line 5 under current CEREMA methodology. The point is therefore not that VAN-ES undercounts welfare by €355–460 million, but that an effect of this magnitude — on the order of construction cost — sits entirely outside the metric that governs the decision, with the fiscal and distributional consequences that follow from it equally invisible.

Furthermore, the pre-construction discount observed in the event study (–€109 to –€149/m² in 2016–2017) suggests that the neighborhoods traversed by Line 5 were undervalued before the project became credible. The tramway investment thus generated a double benefit: improving accessibility and reversing a pre-existing real estate discount in less-favored neighbourhoods. This distributional dimension — which benefits lower-income areas disproportionately, as Section 4.4 demonstrates quantitatively — is entirely absent from current VAN-ES methodology and is worth highlighting to municipal decision-makers.

In the end, a metric that omits a benefit on the order of project cost will systematically under-rank precisely the integrated projects that generate it, and the value such projects could bring to the community.

5.4 The Mode-Relativity of Capitalization: What the Paris T3 Reveals

A natural objection to a mode-based reading of these results comes from Paris. If heavier modes capitalize more strongly, the Tramway des Maréchaux (T3) — a modern, high-capacity tram on dedicated right-of-way — should have produced a substantial premium. It did not. The IAU-IFSTTAR evaluation found that, within 200–400 m of T3 stops, the line's opening was followed by a statistically non-significant price change in the southern inner suburbs (–2% to +7%), and that the immediate frontage of the boulevards des Maréchaux remained a devalued zone, carrying a discount of roughly 5% within 200 m of stops (IAU-IFSTTAR, 2011). On a naïve mode hierarchy, this is anomalous: the same infrastructure that this paper's results showed a premium in Montpellier appears inert in Paris.

The resolution is that capitalization does not depend on the transit mode in absolute terms, but on the marginal accessibility gain the mode delivers relative to the network already in place. A new tram is valuable to a household only insofar as it improves on the options it already has and offers enough

capacity to allow for comfortable travel. In Paris, the T3 was inserted into one of the densest public-transport networks in the world: nearly every point along the Maréchaux already sat within walking distance of a métro station, and the regional RER provided rapid radial access on top of it. The T3's contribution was a tangential surface connection of real operational value — ease traffic intensity on existing network — but its marginal improvement to any given address's accessibility was small since the baseline was already very high. With little accessibility scarcity to relieve and no urban redevelopment alongside it, there was little for the land market to capitalize — exactly the flat or negative pattern the IAU-IFSTTAR study records.

Montpellier sits at the opposite end of this spectrum, and the comparison is best understood by transposing modes across cities rather than comparing them within one. Functionally, the Montpellier tramway plays the role that the métro plays in Paris: it is the structural backbone of the city's rapid-transit network, the mode around which residential location decisions organize. A new tram line in Montpellier therefore delivers the kind of step-change in accessibility that a new métro line — not a new tram — would deliver in Paris, because it expands the primary network rather than supplementing an already-saturated one. By the same transposition, the Bustram BHNS occupies, in Montpellier, the niche the tram occupies in Paris: a useful surface complement layered onto an existing rapid-transit spine, improving connectivity at the margin but rarely transforming a location's accessibility outright. Read this way, the apparent contradiction dissolves. The Montpellier tram and the Paris tram differ in their price effects not because trams behave differently in the two cities, but because the same nominal mode occupies a different rung in each city's accessibility hierarchy. What the land market prices is the rung, not the rolling stock.

This reframing carries directly into the VAN-ES parameter discussion of Sections 5.2 and 5.5. It implies that a mode-specific real estate coefficient, while a clear improvement on a uniform rate, is still incomplete if calibrated on mode alone. The same mode will generate different capitalization in different network contexts: a tram extension in a city where the tram is the backbone is not the same economic object as a tram extension in a city already served by métro and RER. The parameter that travels across projects is therefore not "tram" or "BHNS" as such, but the marginal accessibility gain the project delivers net of the incumbent network. This is precisely the quantity the second refinement in Section 5.6 (calibrating to the marginal accessibility improvement) is designed to capture, and which the Bustram Phase 1 result, layered onto a corridor with substantial existing connectivity, already illustrates in the Montpellier data itself.

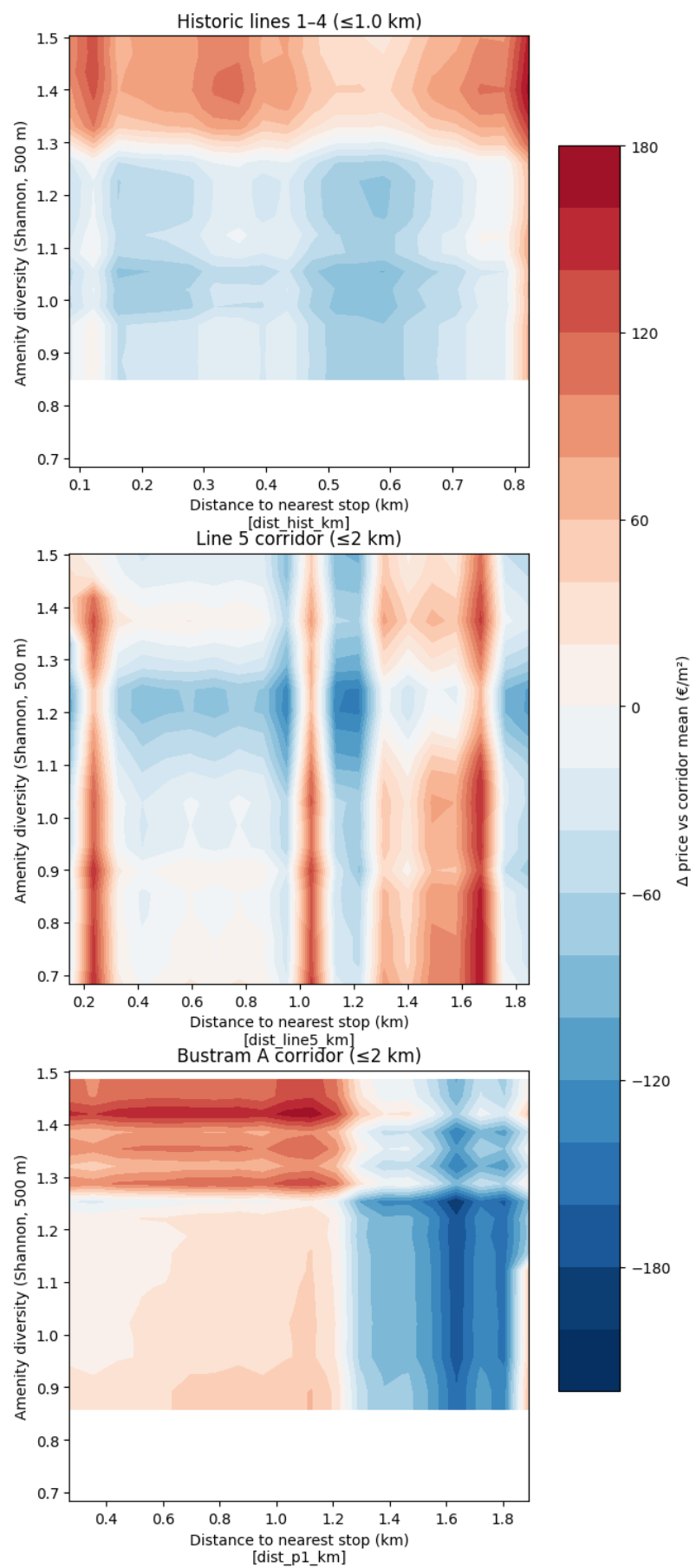
A final caveat must temper the Bustram side of this argument, and it qualifies the mode-relativity reading rather than contradicting it. The present data cannot cleanly separate the price effect of the

Bustram itself from that of the redevelopment accompanying it along the corridor. Bustram capitalization, where detected, is therefore better read as the effect of the transit-plus-redevelopment package than of the mode in isolation. In fact, it is the same identification limit that surfaced for Line 5, whose western alignment coincided with the Cité Créative and Ovalie operations, and it reinforces — rather than undermines — the central policy claim. If the capitalization that materializes around a transit investment is jointly produced by accessibility and the coordinated redevelopment that accompanies it, then the marginal accessibility gain and the planning context are not separable inputs to be estimated independently, but a single bundled treatment. That is exactly why a VAN-ES reform calibrated on mode alone is insufficient, and why the integrated-planning framing of Section 5.6 and the GBM evidence of Section 5.5 — that capitalization is strongest precisely where the amenity and redevelopment context is richest — is the more faithful representation of how transit value is actually created.

5.5 Beyond Causal Effects: Transit, Amenities, and the Capitalization Lifecycle

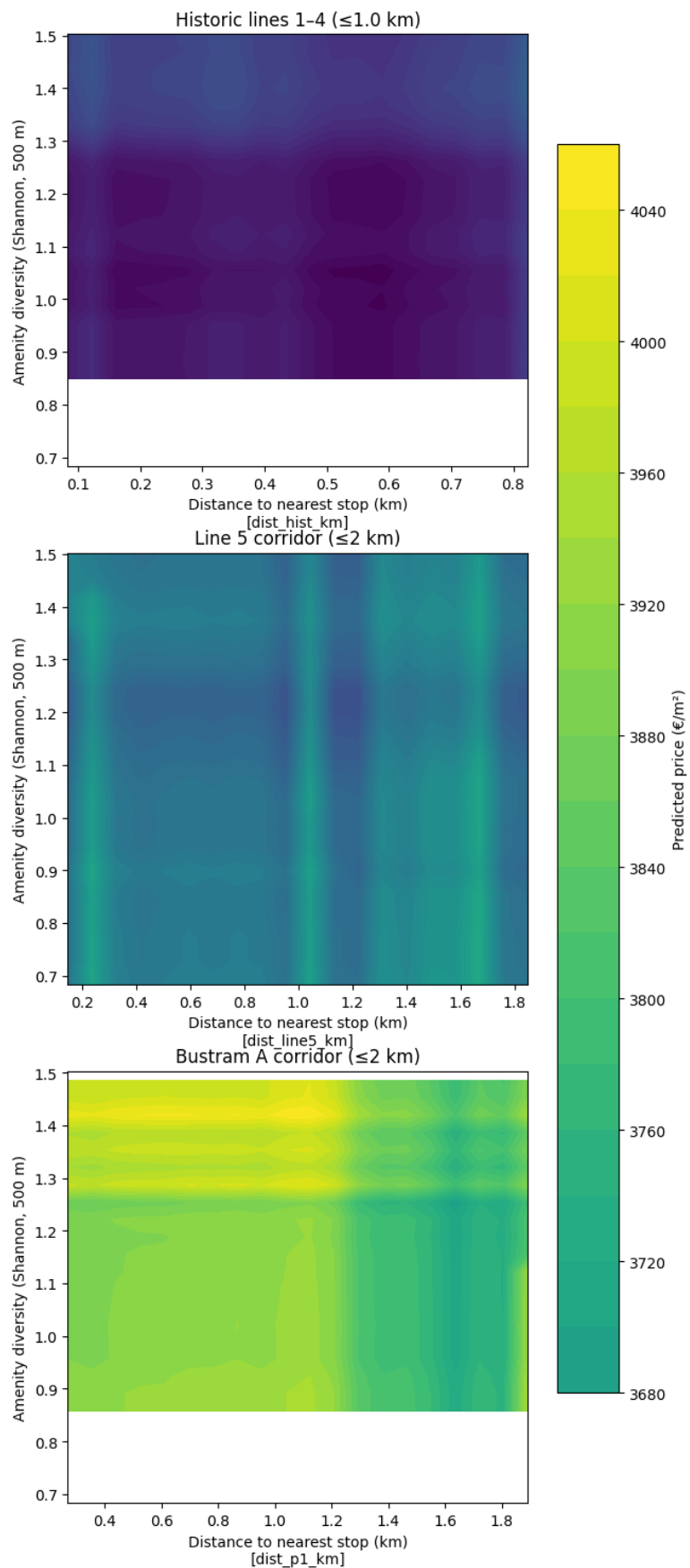
The difference-in-differences results presented in Section 4 identify the causal effect of Line 5's construction on nearby property values and document a smaller, later-arriving effect for Bustram A. These estimates establish that capitalization occurs but say little about the spatial structure in which transit access and the surrounding urban fabric jointly determines prices, or about how this structure evolves as a transit corridor matures. To explore these descriptive dimensions — and to illustrate why incorporating real estate effects into VAN-ES would create incentives for coordinated urban planning rather than purely transit-focused investment — I estimated a complementary gradient-boosted regression model on the same DVF data, augmented with georeferenced amenity points from Montpellier Méditerranée Métropole's open data portal (cafés and bars, restaurants, large retailers, daily-needs shops, services, and marketplaces) and from OpenStreetMap. The detailed methodology for this model can be found as appendix to the paper. The model predicts log price per square metre as a function of property characteristics, postal code, year, and a set of accessibility features measuring distance to each transit mode and amenity density at multiple radius. Because the model is non-parametric and tree-based, it allows non-linear interactions between transit access and the amenity environment, features that a linear hedonic specification cannot capture. The intent is descriptive, not causal — since the DiD has already done the causal work for the construction-credibility effect — the GBM serves to characterize the broader (transit × amenity) price surface in which that effect is embedded.

Figure 22. Within-corridor capitalization gradient



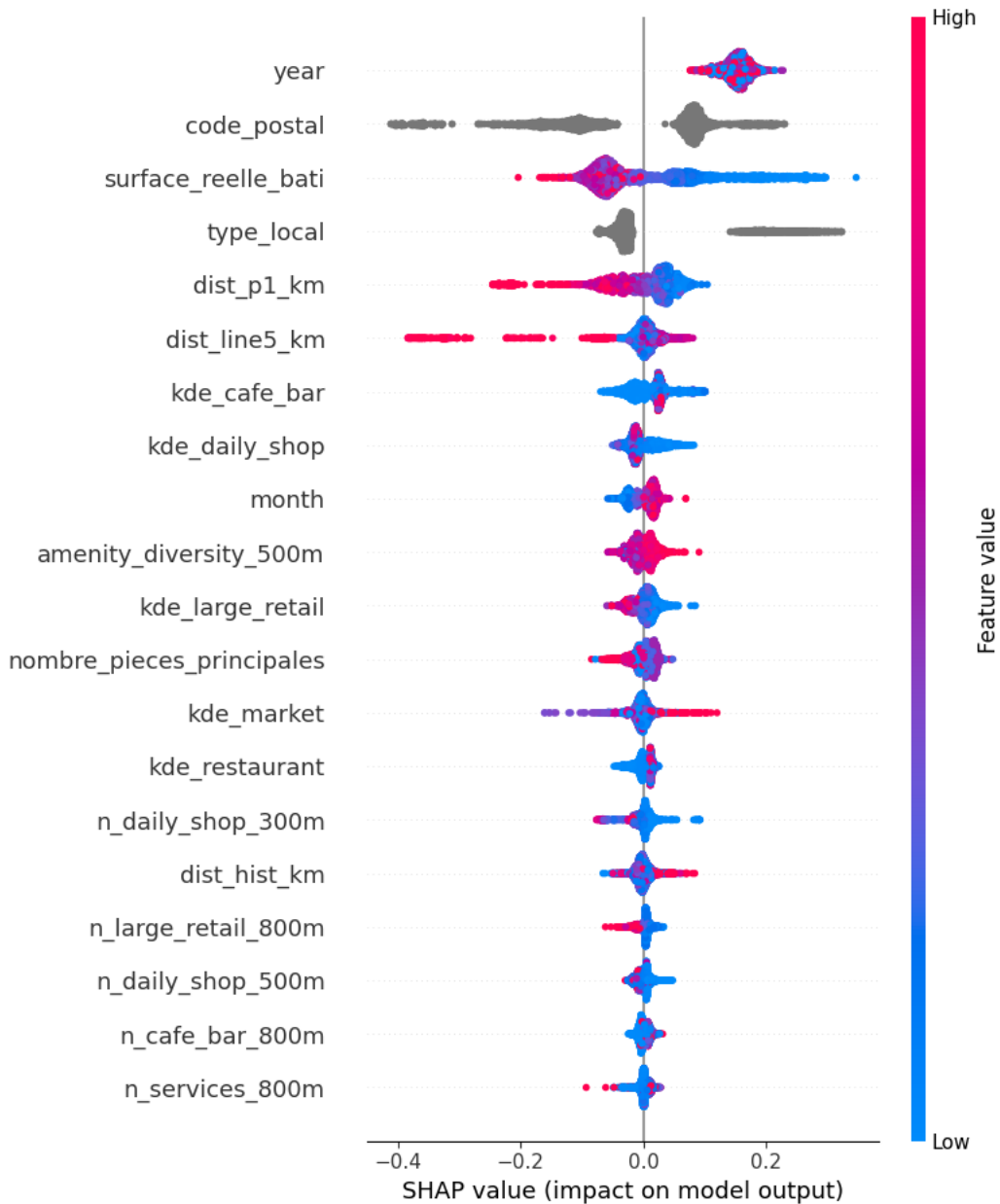
Source: Paul Castet, Code_GBM.

Figure 23. Transit × amenity interaction by mode



Source: Paul Castet, Code_GBM.

Figure 24. SHapeley Additive exPlanation for different variables



Source: Paul Castet, Code_GBM.

Three findings emerge from the analysis. First, the SHAP summary plot (Figure 24) — SHAP, for SHapley Additive exPlanations, being a game-theoretic method that attributes a model's prediction to its individual features (Lundberg & Lee, 2017) — establishes that the amenity environment is a first-order determinant of price per square meter, not a marginal control. The strongest predictors are, as expected, the structural ones: the year of sale, the postal code, built surface, and dwelling type. The two transit-distance variables follow immediately (distance to a Bustram A stop and to a Line 5 stop, ranked fifth and sixth), and the leading amenity measures come directly after them: kernel-density measures of café-and-bar and daily-shop provision, together with the 500 m amenity-diversity index, occupy the seventh through eleventh positions. The amenity environment thus carries explanatory

power of the same order of magnitude as transit access itself — its leading features rank just below the distance-to-stop variables and well above the property's own room count. Two patterns within this ranking are worth noting. The smoothed kernel-density amenity measures consistently outrank their raw buffer-count counterparts (the $n_{\dots}_{300m/500m/800m}$ features cluster at the bottom of the plot), indicating that a continuous representation of the amenity field captures the price signal better than discrete counts within fixed radii. And distance to a historic-line stop ranks far lower than distance to either Line 5 or Bustram A — a first hint, visible in the global importance ranking itself, of the corridor ordering developed below.

Second, partial dependence surfaces estimated on three separate corridor sub-samples — properties within 1 km of a historic-line stop (lines 1–4, in service since 2000–2012), within 2 km of a Line 5 stop, and within 2 km of a Bustram A Phase 1 stop — display a systematic ordering once each surface is normalised to remove between-corridor composition effects. The historic-line corridor exhibits a nearly flat capitalization gradient (within-corridor variation around the corridor mean of approximately $\pm 60 \text{ €/m}^2$): proximity to a historic-line stop barely moves predicted prices conditional on postal-code fixed effects. The Line 5 corridor shows a moderate, somewhat noisier gradient. The Bustram A corridor displays the strongest amenity-conditional pattern ($\pm 180 \text{ €/m}^2$), with a pronounced premium concentrated in the high-amenity-diversity, low-distance corner of the surface. Third, the SHAP interaction value between distance-to-stop and amenity diversity is approximately twice as large in the Bustram A corridor as in the Line 5 corridor, consistent with the visual interpretation of the partial dependence surfaces.

This ordering hints at a coherent interpretation grounded in the temporal dynamics of transit capitalization documented in the empirical literature (Bowes & Ihlanfeldt, 2001; McMillen & McDonald, 2004) and the spatial equilibrium framework of Alonso (1964) and Muth (1969). When transit infrastructure has been operational for sufficient time that housing supply, land use, and commercial activity have fully responded, the marginal capitalization signal flattens to zero as the access premium has been absorbed into the baseline price level of the entire served catchment and into the co-evolved amenity environment (Bowes & Ihlanfeldt, 2001; McMillen & McDonald, 2004). The flat gradient observed for Montpellier's historic tram lines is therefore not evidence of an absence of capitalization; it is evidence of its completion. The mid-cycle Line 5 corridor displays a partially structured gradient consistent with active ongoing repricing, the same process the DiD captures as a $+\text{€}174/\text{m}^2$ treatment effect. The pre-equilibrium Bustram A corridor displays the strongest amenity-conditional pattern because capitalization is still contingent on whether the surrounding urban fabric makes the access gain legibly attractive to buyers. In other words, the accessibility benefit of a young, lower-capacity transit mode has not yet acquired the autonomous valuation that mature heavy-rail

infrastructure commands.

5.6 Implications for VAN-ES Methodology

The results converge on a practical conclusion: property value capitalization is not a granted consequence of transit investment. It depends on the mode, on the marginal accessibility gain the project delivers relative to the existing network, on the credibility of the commitment, and on the planning context surrounding the line. A real estate parameter that ignored these dependencies would misprice projects as surely as the current omission does. I therefore propose four refinements — the first three concerning how the parameter is to be calibrated, the fourth concerning what the reform ultimate intent is.

First, the parameter should be mode-specific and estimated empirically. Both the literature and the results of this thesis indicate that lighter modes (BHNS, standard bus) capitalize less strongly than heavier modes (tramway, metro); a uniform appreciation coefficient would overstate the benefits of bus-based systems and understate the comparative advantage of rail. Such a parameter can be estimated directly with hedonic methods on DVF transaction data, now available for the whole of France, making national calibration feasible rather than hypothetical.

Second, the parameter cannot be a function of mode alone. As the Paris T3 comparison in Section 5.4 shows, the same nominal mode capitalizes differently depending on its position in the host city's accessibility hierarchy. For instance, a tramway forming the backbone of a mid-sized network is not the same economic object as a tramway designed to relieve a saturated métro and RER system. The quantity that travels across projects is the marginal accessibility gain net of the incumbent network, of which mode is only a proxy. Infrastructure that largely duplicates existing connectivity — as parts of Bustram A Phase 1 does — should not be expected to capitalize like infrastructure opening a genuinely new corridor. This does not mean such projects warrant a zero real estate benefit; their evaluation should also capture the improvements they confer on the existing network, since a new route can raise service quality across connecting lines in ways a single-corridor analysis misses and still create the opportunity to have broader urban redevelopment — such as eureka pole along Bustram A Phase 1.

Third, the timing of the parameter must be calibrated to the credibility-establishing event, not to administrative convention. For projects with straightforward political trajectories, the announcement date may serve as a reasonable treatment date, since the market may treat the announcement as a credible commitment. But for projects with contested histories — suspension, relaunch, route changes, or unresolved financing — the announcement-date convention both attenuates the

estimated effect, by averaging over years in which the market had not yet repriced, and overstates the speed of the response, by attributing later capitalization to an earlier signal. The relevant moment is the point at which the project becomes credible enough to trigger sustained repricing; for Line 5 this was the start of irreversible construction, but for other projects it might be the DUP, the financing agreement or the procurement award — and, as the A69 and LGV Bordeaux-Toulouse cases in Section 5.2 suggest, even the DUP may no longer reliably mark that moment. The corollary for practice is that ex-post evaluations measuring outcomes around the completion date will systematically miss gains that materialized years earlier: for Line 5, the bulk of capitalization occurred between 2019 and 2022, well before the December 2025 opening. Evaluation frameworks should accordingly adopt a longer and earlier measurement window, anchored to the credibility event rather than the ribbon-cutting date.

Fourth, — and this is the refinement that gives the reform its purpose — the parameter should be conditioned on the planning environment, because that is the mechanism through which VAN-ES reform becomes an instrument of better urban policy rather than barely a more complete accounting methodology. The descriptive evidence of Section 5.5 shows capitalization is largest where the transit investment sits in a rich, coordinated amenity and redevelopment context, and weakest — for lower-capacity modes especially — where it does not. A VAN-ES modelling capitalization as a function of both mode and planning context would, in doing so, create a quantifiable incentive for coordinated investment: projects pairing a transit line with supportive land-use planning — density allowances near stations, public-realm improvements, retail and service co-location — would score more favorably precisely because the data show capitalization is largest where these conditions are met. This is how the integrated-planning prescription of Higgins and Kanaroglou (2016) — that transit delivers its full potential only alongside complementary land-use coordination — would translate from a descriptive regularity into a concrete budget-evaluation discipline.

The same logic supplies a financing instrument. The demonstrated uplift — on the order of the line's €440 million construction cost in residential value for Line 5 alone (with a transaction-based lower bound of about €161 million) — provides a fiscal basis for value capture, such as a *taxe d'aménagement* modulated at enhanced rates within defined transit corridors; even a partial capture of this increment would materially improve the financial viability of future projects and reduce the burden on public budgets. Beyond financing, a corridor-specific mechanism could itself become an instrument of active redevelopment. The municipality might, for instance, acquire land within the corridor at market rate — itself elevated by the transit premium — and modulate the levy to differentiate between owners: those who sell or redevelop in conformity with the municipality's density objectives would face a reduced levy, while those who retain low-density holdings in a high-accessibility corridor, benefiting

from publicly created appreciation without contributing to its productive use, would face a higher one. Such a graduated structure would align private incentives with the planning objectives that, per Higgins and Kanaroglou (2016), generate transit's largest social returns, and would address a longstanding criticism of value capture — that it can deter beneficial development near transit — by targeting the levy at passive land retention rather than active redevelopment (Smolka, 2013).

Finally, the reform belongs to a broader movement in the measurement frameworks that govern decision-making under externalities. The IFRS S1 and S2 sustainability disclosure standards introduced by the International Sustainability Standards Board in 2023, and the European Union's Corporate Sustainability Reporting Directive, both rest on the recognition that traditional financial accounting systematically excluded effects — climate exposure, environmental impact, social outcomes — that materially shape the welfare consequences of corporate decisions. The reform logic is structurally identical to the one developed here: extend the perimeter of an established metric to capture excluded effects whose omission distorts the choices the metric is meant to inform, building alongside the existing framework rather than replacing it, and accepting that the principal value of the reform lies less in the precision of any single estimate than in the institutional coordination it incentivizes (Adams & Abhayawansa, 2022; Christensen, Hail & Leuz, 2021). IFRS standards operate at the level of the firm and VAN-ES at the level of metropolitan social welfare; the parallel is imperfect, but the institutional insight is shared. Incorporating real estate appreciation into VAN-ES is, in this sense, the transit-investment analogue of a wider effort to align measurement with the full perimeter of consequences that public decisions generate.

5.7 Limitations and future work

Several limitations qualify the interpretation of these results. These limitations are important to acknowledge since they bound what this paper's estimates can and cannot support.

First, the analysis cannot directly identify the capitalization effect of Line 5's opening. The line entered service on 20 December 2025, eleven days before the end of the DVF observation window, and given the standard publication lag of geo-DVF — transactions typically appear three to six months after notarization —, thus no meaningful post-opening sample exists in the present data. The Line 5 estimates therefore identify a construction-credibility effect — the price response to the visible commitment of public works — rather than the equilibrium response to realized accessibility gains. Since the international literature documents that anticipation accounts for a substantial share of total capitalization and often materializes well before opening (Bowes & Ihlanfeldt, 2001; Gibbons & Machin, 2005; Mohammad et al., 2013), a full opening-effect estimate, and an account of convergence

toward equilibrium, would require extending the dataset to at least mid-2026 and ideally to 2027.

Second, the April 2019 construction-start cutoff is applied uniformly across the corridor, whereas the construction in fact proceeded in two phases: works on the northern segment began on 15 April 2019, but works on the western segment — Estanove, Pas-du-Loup, Ovalie, Val-de-Croze — began only on 25 October 2021. Properties along the western branch are thus coded as treated from April 2019 despite no visible construction signal reaching them for a further two and a half years. Under heterogeneous treatment timing, the static two-way fixed-effects estimator delivers a variance-weighted average of cohort-specific effects with potentially negative weights and "forbidden" comparisons between already-treated and newly-treated units (Goodman-Bacon, 2021; de Chaisemartin & d'Haultfœuille, 2020), and the direction of the resulting bias depends on the relative magnitudes of the northern and western effects, which the present specification cannot recover. A staggered design using stop-specific cutoffs, would address this, and I identify it as the most consequential remaining methodological issue. That been said, such analysis would require, to be complete, a qualitative study to correctly frame the impact on real estate professionals and stakeholders on the importance of the announcement effect, the construction signal and the ribbon-cutting.

Third, the Bustram A Phase 1 and Line 5 corridors overlap geographically: they share the postal codes 34000, 34070, 34080, 34090 and 34170. The overlap cuts both ways. Part of the Bustram treated group — the central section of Line A through Antigone and Place de l'Europe — lies within Line 5's zone of influence, so a Bustram premium there may partly capture Line 5 capitalization. More consequentially, a substantial share of the Bustram control group (properties beyond 1 km of a Bustram stop) falls within 500 m of a Line 5 stop — 6,745 properties, or 29% of the corridor — so Line 5 appreciation raises control-group prices and mechanically attenuates the estimated Bustram effect. To address both, the Bustram regressions include the distance to the nearest Line 5 stop and a binary near-Line-5 indicator as covariates. These partial out the time-invariant level difference between near-Line-5 and other properties, but — because they do not interact with the Bustram post-period indicators — they do not absorb the time profile of Line 5 capitalization, so the Bustram coefficients may still reflect, in part, Line 5 dynamics priced in over the same window. A fuller correction would restrict the sample to properties more than 800 m from any Line 5 stop. I implemented the level-difference control but deliberately did not pursue the stronger dynamic correction. The reason is one of purpose rather than convenience: the residual contamination is second-order relative to the aim of this thesis, which is not to isolate a single Bustram coefficient but to demonstrate that real estate capitalization belongs in VAN-ES — and, as Section 5.4 argues, a transit project's value is in any case best understood in the full context of the network and redevelopment surrounding it, not as an artificially isolated shock.

Taken together, these limitations imply that the Line 5 coefficients should be read as construction-credibility effects rather than completion effects, that their magnitude is potentially attenuated by the uniform-timing assumption, and that the Bustram A coefficients estimate a joint response to Line A and residual Line 5 dynamics. None reverses the qualitative direction of the findings, but each constrains the precision with which a euros-per-square-meter figure can be attributed to a single isolated shock. The estimates should therefore be read not as precise valuations but as evidence of an order of magnitude — sufficient to establish that the omitted capitalization is large enough to matter for VAN-ES, which is the claim the thesis rests on.

A further set of limitations applies specifically to the gradient-boosted model of Section 5.5 and confines it to a descriptive role. Most fundamentally, the GBM does not establish that amenity environments cause stronger capitalization; it documents only that, in the Montpellier cross-section, the two are jointly distributed in a pattern consistent with the spatial-equilibrium prediction. Quasi-experimental identification of the amenity–transit interaction would require independent variation in amenity environments — for instance, a comparison of two corridors with identical transit infrastructure but differing densification trajectories — which lies beyond the present data and is left to future work.

A second, more technical limitation concerns the dependent variable. Price per square metre is computed as `valeur_fonciere / surface_reelle_bati`, normalizing transaction price by interior built area for both apartments and houses. This does not adjust for the land share embedded in the transaction: for houses the land allocation is the parcel itself, while for apartments it depends on the building's floor count and unit density. The estimated effects therefore reflect a composite of land value and structure value rather than land value alone, even though transit capitalization is theoretically a land-price phenomenon (Alonso, 1964; Mills, 1972); the property-type fixed effect absorbs only the average level difference between the two regimes, not the within-type variation in land intensity, introducing an attenuation bias that differs across property types. A more rigorous treatment would derive a true land-price target — direct for houses (`valeur_fonciere / surface_terrain`), and for apartments via the building footprint and the unit count in the cadastral parcel to obtain a per-unit land quote-part. The CEREMA Fichiers Fonciers dataset, supplies both the building geometry and the residential unit count per building and would enable this decomposition. Additionally, the recent work of IGN on density maps using deep learning models could help with figuring out the evolution of density to better reflect any density premium occurring. Re-estimating on decomposed land prices would clarify whether the lower Bustram A premium reflects genuinely weaker capitalization or simply higher building density along the BHNS corridor, and would provide a more theoretically grounded input to the VAN-ES uplift

calculation, which is itself a land-value framework.

Finally, the conclusions drawn for the BHNS mode are necessarily preliminary. The Phase 1 analysis rests on a single Bustram line in a corridor with substantial overlapping connectivity, and the Phase 2 analysis on a smaller sample identified off an announcement alone. The planned extension to Castries — through communes with far less developed transit networks — will eventually offer a stronger test of whether a BHNS, despite its "tram" branding, can generate property value effects where the marginal accessibility gain is genuinely large. The econometric framework developed here, and the attached scripts are ready to be re-applied once DVF data become available for 2026 and beyond.

6. Conclusion

This thesis asked whether real estate appreciation should be incorporated into the VAN-ES socio-economic evaluation framework used to appraise major infrastructure in France. The conclusion is affirmative — but the evidence attaches conditions to how such an extension should be designed, and those conditions matter as much as the principle.

The empirical core rests on a difference-in-differences analysis of two contemporaneous transit projects in the same metropolitan area, Montpellier: tramway Line 5 and the Bustram A Bus à haut niveau de service (BHNS), studied with a common methodology. Using DVF transaction data covering 2015-2025 and a donut treatment-control design with property-level controls, postal-code fixed effects, and seasonal indicators, the preferred specification estimates that properties within 500 meters of a Line 5 station appreciated by roughly €174/m² — about 4.58% — more than control properties beyond one kilometer once construction began in April 2019. The estimate is stable in sign, significance, and order of magnitude across heteroskedasticity-consistent and cluster-robust standard errors, across the level and log specifications, and across the choice of estimation window. Scaled to the residential stock within 500 metres of a station, the implied capitalization attributable to Line 5 is on the order of the project's €440 million construction cost — a transaction-based lower bound puts it at about €161 million (~37% of cost) — a material magnitude entirely absent from the VAN-ES calculation that supported the project, and one that, under capitalization equivalence, is best read as a scale marker rather than a clean addition to the benefit side.

The Bustram A analysis qualifies and enriches this finding. Phase 1, the operating line, generates a smaller and later-arriving effect of roughly €101/m² ($p = 0.031$), detectable only at the construction-cutoff specification and absent at the announcement, relaunch, and opening cutoffs. Phase 2, the planned Castries extension, returns a noisy negative coefficient at the announcement cutoff that should be read as a compositional artefact rather than a genuine market response, given the small treatment group and the absence of any construction-stage shock to benchmark against. Read together, the estimates trace a credibility ladder: a heavy, costly, irreversible tramway under construction commands the largest premium; a lighter but visible BHNS construction work moves prices by roughly half as much; a project known only through political announcement does not move them measurably. Mode matters but what matters more is the credibility of the public commitment that the mode and the project stage convey to the market.

This credibility-based reading of when capitalization occurs is the thesis's primary methodological contribution. The standard "announcement effect" documented in the literature (Gibbons & Machin,

2005; McMillen & McDonald, 2004) rests on projects with stable political trajectories. For projects with contested histories — suspension, relaunch, route changes, uncertain financing — the relevant treatment date is not the formal announcement but the moment at which physical construction renders the project credible and effectively irreversible. The Line 5 event study, with no break at the 2016 relaunch and a clean structural break at the 2019 construction start, illustrates the distinction sharply; recent French infrastructure history — the judicial cancellation of the A69, the unresolved financing of the LGV Bordeaux–Toulouse — suggests it is becoming more important, not less. Empirical work on transit capitalization in France should therefore identify the credibility-establishing event from the data rather than defaulting to an administrative milestone.

A second contribution concerns distribution. The heterogeneous-effects analysis shows that capitalization in lower-price postal codes — those traversing the *quartiers prioritaires de la politique de la ville* of western Montpellier — was roughly twice the premium in higher-price areas, with a total effect approaching €210/m² in low-price corridors against €106/m² in high-price ones. This is consistent with the prediction that marginal accessibility gains are valued most where pre-existing connectivity is weakest (Kahn, 2007), and it bears directly on the equity-weighted parameters that Quinet (2013) recommends for socio-economic evaluation: a distributional weight in an extended VAN-ES would further raise the computed social value of transit that serves disadvantaged neighbourhoods.

The descriptive gradient-boosted analysis of the joint transit × amenity price surface reinforces the point from another angle: capitalization is a multi-year process whose magnitude depends not only on the transit investment but on the amenity and redevelopment environment that grows around it. An evaluation framework that quantifies real estate effects would therefore reward not transit investment in isolation, but transit investment coordinated with supportive land-use planning — precisely the condition under which the international evidence (Higgins & Kanaroglou, 2016) finds that transit delivers its full potential.

The proposal that emerges is accordingly not a mechanical add-on but a structured extension. Real estate parameters should be calibrated empirically and differentiated by mode — though, as the Paris T3 comparison shows, mode is only a proxy for the deeper quantity, the marginal accessibility gain a project delivers relative to the network already in place. The treatment date should be the credibility-establishing event, identified from the data rather than by convention. The estimate should accommodate distributional weighting where the corridor runs through areas of socioeconomic disparity. And the framework should be paired with appropriate value-capture mechanisms — a modulated *taxe d'aménagement*, targeted developer participations, or a Section 106-style instrument

— through which the publicly created uplift could help finance the infrastructure that generated it and even create incentives to redevelop land more coherently, while giving decision-makers and the public a clearer, more transparent account of a project's financing options. The fiscal magnitude of the Line 5 uplift suggests that even small capture would materially improve the viability of comparable future projects and ease the burden on stretched municipal budgets; the same mechanisms could also steer private investment toward more comprehensive redevelopment of a corridor.

Several limitations qualify these conclusions and are discussed in detail in Section 5.7. The Line 5 estimates identify a construction-credibility effect rather than a completion effect, since DVF coverage of post-opening transactions is mechanically incomplete as of writing. The uniform April 2019 treatment date abstracts from the staggered construction timing of the northern and western sections. The Bustram A specifications partial out the Line 5 level effect but not its time profile, leaving open the possibility that the Bustram coefficients reflect a joint response to both projects. The price-per-built-square-meter normalization conflates land and structure value in a way the underlying land-price theory does not.

Two further limitations bear directly on the central interpretation and define the most valuable next steps. First, this thesis cannot fully disentangle what moves prices — the credibility of the public commitment, or the comprehensive redevelopment that the commitment sets in motion. Line 5's western alignment coincided with the Cité Créative and Ovalie operations, and the Bustram corridor was likewise accompanied by redevelopment around Eureka pôle, so the estimated premia are best read as the effect of a transit-plus-redevelopment package rather than of accessibility alone. A quantitative design cannot separate these channels where they are bundled in time and space; a qualitative complement — interviews with developers, notaires, local planners, and buyers along the corridor — would be needed to establish which signal actually drives the repricing, and is a priority for future work. Second, none of these issues reverses the qualitative direction of the findings, but each constrains the precision with which a euros-per-square-meter figure can be attributed to a single isolated shock; the estimates should be read as evidence of an order of magnitude, not as exact valuations. The natural next steps follow directly: extend the DVF window into 2026 and beyond to recover a genuine opening effect and trace convergence toward equilibrium; apply the framework across other French urban settings rather than Montpellier alone; re-estimate on the Castries extension once construction-period data exist; and decompose the price target into land and structure components using CEREMA's Fichiers Fonciers dataset.

The broader point I want to leave with the reader is that measurement frameworks shape the decisions they measure. Excluding real estate appreciation from VAN-ES is not a neutral methodological choice:

it systematically undervalues transit whose primary benefit is urban transformation rather than travel-time savings, and it offers no quantifiable incentive for the integrated planning that delivers transit's full potential. And crucially, omitting the effect does not make it disappear — the land market reprices around credible infrastructure regardless of what the evaluation framework records. What omission changes is not the outcome but our capacity to anticipate and govern it: an appreciation that goes unmeasured is harder to foresee, harder to plan for, and harder to deliberate — including the displacement pressure it places on existing residents and the corresponding need for compensatory measures such as social housing in the very neighborhoods the project most benefits. Quantifying the effect explicitly would make the value a public project creates — and thus the question of who captures it — an object of deliberation at the *enquête publique* rather than an unmeasured by-product. The contemporary movement toward expanded measurement frameworks — IFRS S1 and S2, the EU's Corporate Sustainability Reporting Directive, the broader recognition that excluded externalities do not shrink by going unmeasured — is structurally analogous to the reform proposed here. Extending VAN-ES to capture real estate effects is not a refinement to a marginal parameter; it is an alignment of the metric with the full perimeter of consequences that public investment in urban transit generates. The case of Line 5 — residential capitalization on the order of the project's construction cost, concentrated in the neighborhoods that needed it most, and entirely absent from the calculation that sanctioned the project — illustrates both the size of the omission and the case for closing it.

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Appendix A — Gradient Boosted Hedonic Model: Specification and Implementation

A.1 Purpose and modelling choice

The descriptive analysis in Section 5.4 uses a gradient boosted regression model rather than a linear hedonic specification. The choice is driven by the analytical objective: characterising the joint structure of the (transit access $\times$ amenity environment) price surface, including potential non-linear interactions that a linear specification cannot accommodate. Gradient boosting is well-suited to this task — it accommodates non-linearities and high-order interactions without functional-form assumptions, handles mixed numeric/categorical features natively, and produces SHAP decompositions that allow per-feature attribution of predictions. The model is intended as descriptive and complementary to the causal DiD analysis of Section 4; it does not identify a treatment effect and is not interpreted as such.

A.2 Sample and target variable

The model is estimated on the same DVF transactions used for the Line 5 analysis, applying the cleaning conventions described in Section 3.2 (sale transactions only, apartments and houses, price per square metre between €500 and €10,000, year 2020 excluded), augmented with the four communes of the Bustram A Phase 2 corridor. The target variable is the natural logarithm of price per square metre. Training uses transactions from 2015 to 2022 (excluding 2020); testing uses transactions from 2023 to 2025. This temporal split ensures out-of-sample evaluation is a genuine prediction exercise rather than a within-sample fit.

A.3 Feature set

Thirty-five features are used in three groups. Hedonic controls comprise built surface, number of rooms, property type (apartment / house, as a categorical), postal code (categorical, five levels), year, and month. Transit accessibility is measured as Euclidean distance in kilometres from each transaction to the nearest stop on Line 5, on Bustram A Phase 1, and on the historic-line network (lines 1–4), using the equirectangular projection with cosine-latitude correction described in Section 3.2.2. Amenity environment is measured in two complementary ways: count of amenity points within 300, 500, and 800-metre buffers for each of six categories (cafés/bars, restaurants, large retailers, daily-needs shops, services, and marketplaces), and Gaussian kernel-density intensity at each transaction location with a 400-metre bandwidth. A Shannon entropy index over the six categories at the 500-metre radius captures amenity-mix diversity; a total-amenity count provides the overall density. The six amenity categories are constructed from OpenStreetMap tags through the classification rules described in the notebook (Appendix C, cells 3–4).

A.4 Spatial Amenity Data

The local amenity environment is characterized using point-of-interest data extracted from OpenStreetMap and Opendata for the Montpellier metropole, drawn from three layers (economic establishments, food-and-drink venues, and marketplaces). Each point is classified into one of six categories:

- Restaurants (1,355 points): restaurants and fast-food outlets.
- Daily shops (809 points): bakeries, butchers, greengrocers, convenience stores, newsagents, delis, cheesemongers, fishmongers, and pastry shops — walkable, everyday food retail generally associated with residential desirability.
- Services (783 points): hairdressers, beauty salons, laundries and dry cleaners, tailors, opticians, shoe and key repair, watch shops, and florists — routine personal services.
- Cafés and bars (372 points): cafés, bars, pubs, and ice-cream parlours.
- Large retail (107 points): supermarkets, hypermarkets, malls, department stores, and wholesale outlets — large-format retail typically located in commercial zones.
- Marketplaces (55 points): open-air and covered markets.

For each property and each category, it is possible to construct two families of spatial features. The first is a set of buffer counts — the number of establishments within 0.3 km, 0.5 km, and 0.8 km — capturing amenity density at three spatial scales. The second is a Gaussian kernel-density estimate (bandwidth 0.4 km), a smoother analogue of the buffer counts that is less sensitive to arbitrary radius cut-offs. This yields 24 amenity features in total (six categories × three buffer radii, plus six kernel-density measures). Distance-to-transit variables (to the nearest Line 5 and Bustram stops) are handled separately as the accessibility regressors of interest, and walking-network distances computed via OSMnx are optionally added alongside the straight-line measures.

A.5 Estimation

The model is implemented using LightGBM (Ke et al., 2017) with the following hyperparameters: 2,000 maximum boosting rounds, learning rate 0.03, 63 leaves per tree (31 for the smaller corridor subsamples), minimum 30 observations per leaf, 80% subsampling of rows and columns at each tree, and an L2 regularisation term of 1.0. Early stopping on the validation set with a patience of 50 rounds determines the final number of trees actually used (414 rounds for the pooled metro model; 366–663 for corridor-specific models). The pooled model uses 36,390 training and 15,492 test observations; the Line 5 corridor model 28,950 and 11,773; the Bustram A corridor model 16,194 and 6,971; the historic-

line corridor model (≤ 1 km) 33,688 and 14,337.

A.6 Out-of-sample performance

Out-of-sample R^2 ranges from 0.323 (Bustram A corridor) to 0.402 (Line 5 corridor); median absolute percentage error is 13.1–13.3% across all four models. These figures are comparable to those reported by Fack and Grenet (2010) on similar French DVF data and reflect both the absence of detailed property characteristics in DVF (floor level, energy rating, condition) and the structural extrapolation involved in predicting 2023–2025 prices from 2015–2022 training data — a period spanning the post-COVID rate-cycle transition during which French residential prices underwent material adjustment. The fit is sufficient for the descriptive purpose of the analysis; it is not interpreted as a forecast of individual transaction prices.

A.7 SHAP decomposition and partial dependence

Feature importance and interactions are analysed using SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017), computed with the TreeExplainer algorithm. The summary plot in Figure 22, 22, 23 displays the global ranking; SHAP interaction values reported in Section 5.4 are computed on a 1,500-observation subsample drawn from the test set. Two-dimensional partial dependence surfaces are computed using a manual grid-based implementation rather than the `scikit-learn partial_dependence` function: for each grid point, the corresponding two-feature pair is set to the grid value across a 2,000-observation subsample, the model predicts for each row, and predictions are averaged. The grid uses 25 points per dimension (20 for corridor-specific surfaces) spanning the 5th to 95th percentile of each feature. Surfaces are exponentiated to the €/m² scale for interpretation. Centred surfaces (Figure 5.4.2) subtract the panel-mean prediction from each cell, isolating within-corridor structure from between-corridor composition effects.

A.8 Reproducibility

The complete implementation, including all data-loading, feature-engineering, and figure-generation code, is provided in Appendix C. The pipeline is deterministic conditional on the random seed (42 throughout); the principal sources of stochasticity are the row and column subsampling within LightGBM trees. Re-running the notebook on the same DVF input files reproduces all reported figures.

Code and data availability. All code required to reproduce the results, figures, and tables in this thesis — the difference-in-differences analysis (Code_DiD) and the gradient-boosted hedonic model (Code_GBM) — is openly available at <https://github.com/Montpellier-Tram-thesis/montpellier-van-es-thesis> under an MIT licence. The repository contains the analysis notebooks, an environment specification, and scripts to download the underlying open data. Property-transaction data come from the Demandes de Valeurs Foncières (geo-DVF) registry (Direction Générale des Finances Publiques, Licence Ouverte/Etalab 2.0); tramway and amenity geometries from Montpellier Méditerranée Métropole's open-data portal and from OpenStreetMap (© OpenStreetMap contributors, ODbL); and building-permit counts from the Sit@del2 database. The raw DVF files can be downloaded on data.gouv.fr.